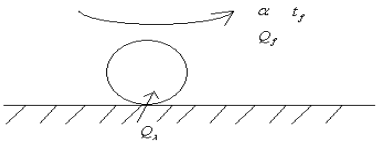
绪论

思考题与习题（P_{8-9}）答案：

1． 冰雹落体后溶化所需热量主要是由以下途径得到：



Q_λ —— 与地面的导热量

Q_f ——与空气的对流换热热量

注：若直接暴露于阳光下可考虑辐射换热，否则可忽略不计。

6． 夏季：在维持 20℃的室内，人体通过与空气的对流换热失去热量，但同时又与外界和内墙面通过辐射换热得到热量，最终的总失热量减少。（$T_{外}\rangle T_{内}$）

冬季：在与夏季相似的条件下，一方面人体通过对流换热失去部分热量，另一方面又与外界和内墙通过辐射换热失去部分热量，最终的总失热量增加。（$T_{外}\langle T_{内}$）

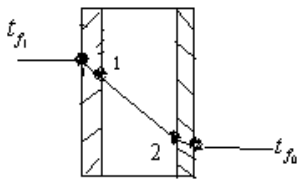
挂上窗帘布阻断了与外界的辐射换热，减少了人体的失热量。

7． 热对流不等于对流换热，对流换热 = 热对流 + 热传导

热对流为基本传热方式，对流换热为非基本传热方式

8． 门窗、墙壁、楼板等等。以热传导和热对流的方式。

9． 因内、外两间为真空，故其间无导热和对流传热，热量仅能通过胆壁传到外界，但夹层两侧均镀锌，其间的系统辐射系数降低，故能较长时间地保持热水的温度。



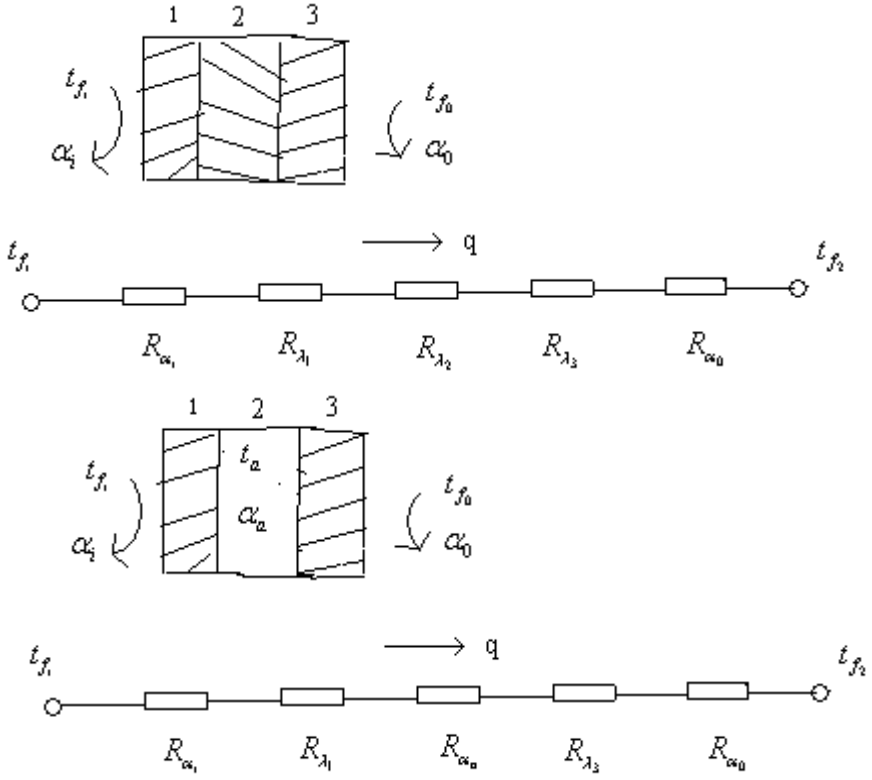
当真空被破坏掉后，1、2 两侧将存在对流换热，使其保温性能变得很差。

10． $R_{\lambda t}=\dfrac{R_\lambda}{A} \quad \Rightarrow \quad \dfrac{R_{\lambda t}}{R_\lambda}=\dfrac{1}{A}=\dfrac{1}{12}=8.33\times10^{-2}m^{-2}$

11． $q=\dfrac{\lambda}{\sigma}\Delta t$ $\qquad \lambda=const \rightarrow$ 直线

$\lambda\neq const$ 而为 $\lambda=\lambda(t)$ 时 $\rightarrow$ 曲线

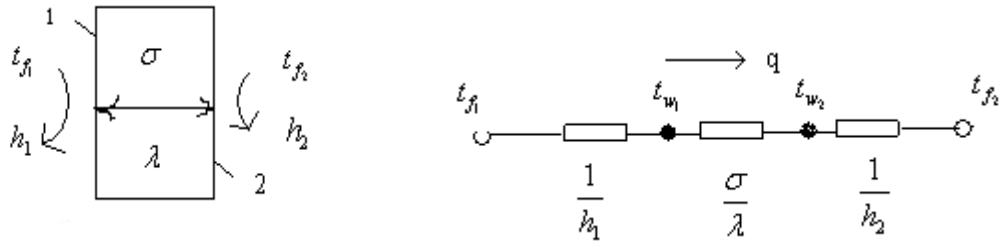
12. R_{α_i} R_{λ_1} R_{λ_3} R_{α_0} t_{f_1} $\longrightarrow$ **q**



首先通过对流换热使炉子内壁温度升高，炉子内壁通过热传导，使内壁温度生高，内壁与空气夹层通过对流换热继续传递热量，空气夹层与外壁间再通过热传导，这样使热量通过空气夹层。（空气夹层的厚度对壁炉的保温性能有影响，影响α_a的大小。）

13. 已知：$\sigma = 360mm$、$\lambda = 0.61 W/(m \bullet K)$　　$t_{f_1} = 18$℃　　$h_1 = 87 W/(m^2 \bullet K)$

$t_{f_2} = -10$℃　　$h_2 = 124 W/(m^2 \bullet K)$　　**墙高 2.8m，宽 3m**



求：q、t_{w_1}、t_{w_2}、ϕ

解：$q = \dfrac{\Delta t}{\dfrac{1}{h_1} + \dfrac{\sigma}{\lambda} + \dfrac{1}{h_2}} = \dfrac{18 - (-10)}{\dfrac{1}{87} + \dfrac{0.36}{0.61} + \dfrac{1}{124}} = 45.92 \, W/m^2$

$q = h_1(t_{f_1} - t_{w_1}) \Rightarrow \quad t_{w_1} = t_{f_1} - \dfrac{q}{h_1} = 18 - \dfrac{37.54}{87} = 17.57$℃

$$q = h_2(t_{w_2} - t_{f_2}) \Rightarrow \quad t_{w_2} = t_{f_2} + \frac{q}{h_2} = -10 + \frac{37.54}{124} = -9.7\ ℃$$

$$\phi = q \times A = 45.92 \times 2.8 \times 3 = 385.73W$$

14. 已知：$H = 3m$、$\sigma = 0.2m$、$L = 2m$、$\lambda = 45\ {}^{W}\!/_{(m \bullet K)}$ $t_{w_1} = 150\ ℃$、$t_{w_2} = 285\ ℃$

求：$R_{\lambda t}$、R_{λ}、q、ϕ

解：$R_{\lambda t} = \frac{\sigma}{\lambda A} = \frac{\sigma}{\lambda HL} = \frac{0.2}{45 \times 3 \times 2} = 7.407 \times 10^{-4}\ {}^{K}\!/_{W}$

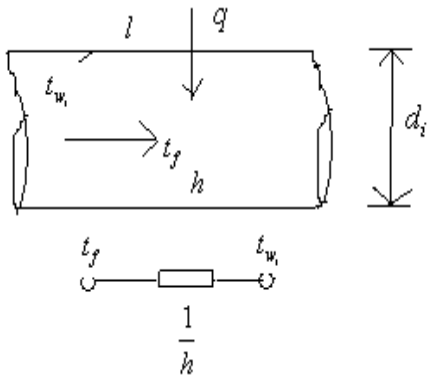
$$R_{\lambda t} = \frac{\sigma}{\lambda} = \frac{0.2}{45} = 4.444 \times 10^{-3}\ {}^{m^2 \bullet K}\!/_{W}$$

$$q = \frac{\Delta t}{R_{\lambda}} = \frac{285 - 150}{4.444 \times 10^{-3}} \times 10^{-3} = 30.4\ {}^{KW}\!/_{m^2}$$

$$\phi = \frac{\Delta t}{R_{\lambda t}} = \frac{285 - 150}{7.407 \times 10^{-4}} \times 10^{-3} = 182.3KW$$

15. 已知：$d_i = 50mm$、$l = 2.5m$、$t_f = 85℃$、$h = 73\ {}^{W}\!/_{(m^2 \bullet K)}$、$q = 5110\ {}^{W}\!/_{m^2}$

求：t_{w_i}、ϕ



$$q = h\Delta t = h(t_{w_i} - t_f)$$

$$\Rightarrow t_{w_i} = t_f + \frac{q}{h}$$

$$= 85 + \frac{5110}{73} = 155\ ℃$$

$$\phi = Aq = \pi d_i l q = 0.05\pi \times 2.5 \times 5110 = 2006.7W$$

16. 已知：$t_{w_1} = 50\ ℃$、$t_{w_2} = 20\ ℃$、$c_{1.2} = 3.96\ {}^{W}\!/_{(m^2 \bullet K^4)}$、$t'_{w_1} = 200\ ℃$

求：$q_{1.2}$、$q'_{1.2}$、$\Delta q_{1.2}$

解：$q_{1.2} = c_{1.2}\left[(\frac{t_{w_1}}{100})^4 - (\frac{t_{w_2}}{100})^4\right]$

$$= 3.96 \times \left[(\frac{273+50}{100})^4 - (\frac{273+20}{100})^4\right] = 139.2\ {}^{W}\!/_{m^2}$$

$$q'_{1.2}=c_{1.2}\left[(\frac{t'_{w_1}}{100})^4-(\frac{t_{w_2}}{100})^4\right]$$

$$=3.96\times\left[(\frac{273+200}{100})^4-(\frac{273+20}{100})^4\right]=1690.3\,W/m^2$$

$$\Delta q_{1.2}=q'_{1.2}-q_{1.2}=1690.3-139.2=1551.1\,W/m^2$$

17．已知：$A=24m^2$、$h_1=5000\,W/(m^2\bullet K)$、$h_2=85\,W/(m^2\bullet K)$、$t_1=45℃$

$t_2=500℃$、$k'=h_2=85\,W/(m^2\bullet K)$、$\sigma=1mm$、$\lambda=398\,W/(m\bullet K)$

求：k、ϕ、Δ

解：由于管壁相对直径而言较小，故可将此圆管壁近似为平壁

即：$k=\dfrac{1}{\dfrac{1}{h_1}+\dfrac{\sigma}{\lambda}+\dfrac{1}{h_2}}=\dfrac{1}{\dfrac{1}{5000}+\dfrac{1\times10^{-3}}{390}+\dfrac{1}{85}}=83.56\,W/(m^2\bullet k)$

$$\phi=kA\Delta t=83.56\times24\times(500-45)\times10^{-3}=912.5KW$$

若$k\approx h_2$

$$\Delta=\frac{k'-k}{k}\times100\%=\frac{85-83.56}{83.56}=1.72\%$$

因为：$\dfrac{1}{h_1}$□ $\dfrac{1}{h_2}$，$\dfrac{\sigma}{\lambda}$□ $\dfrac{1}{h_2}$

即：水侧对流换热热阻及管壁导热热阻远小于燃气侧对流换热热阻，此时前两个热阻均可以忽略不记。

18．略

第一章导热理论基础

思考题与习题（P_{24}）答案：

1．略

2．已知：$\lambda_1=0.62\,W/(m\bullet K)$、$\lambda_2=0.65\,W/(m\bullet K)$、$\lambda_3=0.024\,W/(m\bullet K)$、

$\lambda_4 = 0.016 W/(m \bullet K)$

求： R'_λ、R''_λ

解： $R'_\lambda = \frac{\sigma_1 \times 2}{\lambda_1} + \frac{\sigma_2}{\lambda_2} + \frac{\sigma_4 \times 2}{\lambda_4} = \left(\frac{4 \times 2}{0.62} + \frac{5}{0.65} + \frac{9 \times 2}{0.016} \right) \times 10^{-3} = 1.146 m^2 \bullet K/W$

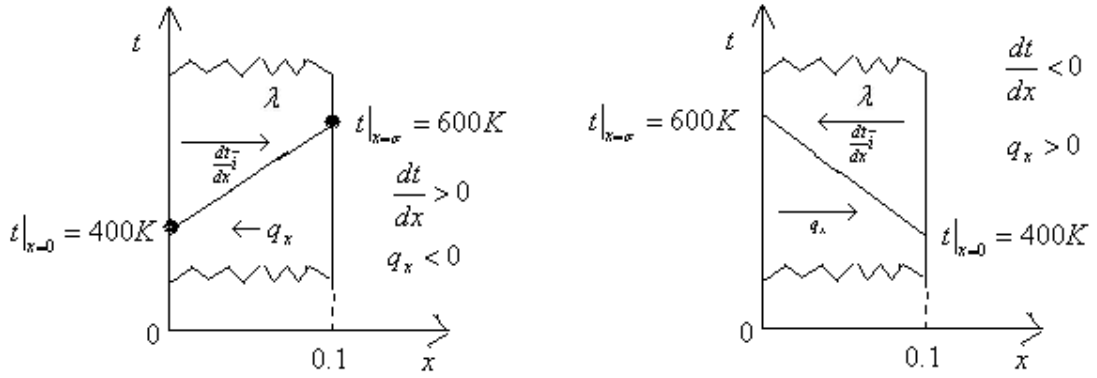
$$R''_\lambda = \frac{\sigma'_2}{\lambda_2} + \frac{\sigma_3}{\lambda_3} = \left(\frac{2 \times 5}{0.65} + \frac{6}{0.024} \right) = 0.265 m^2 \cdot k / W$$

由计算可知，双 Low-e 膜双真空玻璃的导热热阻高于中空玻璃，也就是说双 Low-e 膜双真空玻璃的保温性能要优于中空玻璃。

3.

4．略

5 .



6．已知： $\sigma = 50mm$、$t = a + bx^2$、$a = 200℃$、$b = -2000℃/m^2$、$\lambda = 45 W/(m \bullet K)$

求：（1）$q|_{x=0}$、$q|_{x=6}$ （2）q_v

解：（1）$q|_{x=0} = -\lambda \left. \frac{dt}{dx} \right|_{x=0} = -\lambda 2bx|_{x=0} = 0$

$$q|_{x=\sigma} = -\lambda \left. \frac{dt}{dx} \right|_{x=\sigma} = -\lambda 2bx|_{x=\sigma} = -45 \times 2 \times (-2000) \times 50 \times 10^{-3} = 9 \times 10^3 W/m^2$$

（2）由 $\frac{d^2 t}{dx^2} + \frac{q_v}{\lambda} = 0$

$$q_v = -\lambda \frac{d^2 t}{dx^2} = -\lambda 2b = -45 \times (-2000) \times 2 = 180 \times 10^3 W/m^3$$

7．略

8．略

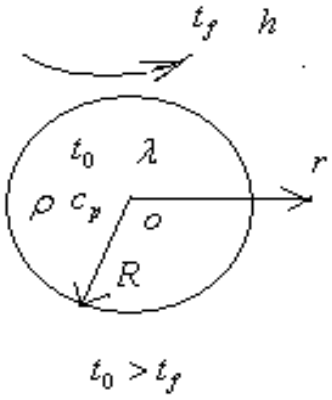
9．取如图所示球坐标，其为无内热源一维非稳态导热

故有：$\frac{\partial t}{\partial \tau}=\frac{a}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial t}{\partial r}\right)$

$\tau=0, t=t_0$

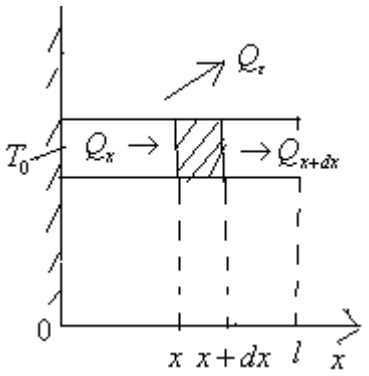
$r=0, \frac{\partial t}{\partial r}=$（

$r=R, -\lambda\frac{\partial t}{\partial r}=h(t-t_f)$



$t_0 > t_f$

10．解：建立如图坐标，在 x=x 位置取 dx 长度微元体，根据能量守恒有：



$Q_{x+dx}+Q_\varepsilon=Q_x$ （1）

$Q_x=-\lambda+\frac{dt}{dx}$ $Q_{x+dx}=-\lambda+\frac{d}{dx}(t+\frac{dt}{dx}\bullet dx)$

$Q_\varepsilon=EA=\varepsilon E_b A=\varepsilon\sigma_b T^4(Udx)$

代入式（1），合并整理得：

$$\frac{d^2t}{dx^2}-\frac{\varepsilon\sigma_b U}{\lambda_f}T^4=0$$

该问题数学描写为：

$$\frac{d^2t}{dx^2}-\frac{\varepsilon\sigma_b U}{\lambda_f}T^4=0$$

$x=0, t=T_0$

$x=l, \left.\frac{dt}{dx}\right|_{x=l}=0$(假设的)

$-\lambda f\left.\frac{dt}{dx}\right|_{x=l}=\varepsilon\sigma_b T_e^4 f$(真实的)

第二章稳态导热

思考题与习题（$P_{51\text{-}53}$）答案

1.略　　2.略

3.解：（1）温度分布为 $t=t_{w1}-\frac{t_{w1}-t_{w2}}{\delta}x$ **(设** $t_{w1}>t_{w2}$ **)**

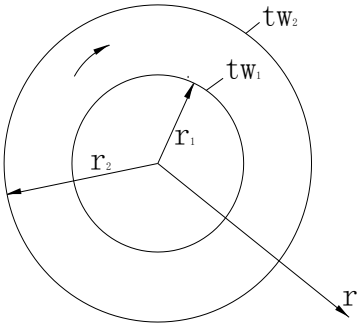
其与平壁的材料无关的根本原因在 $\lambda=cous$（即常物性假设），否则 t 与平壁的材料有关

（2）由 $q=-\lambda\frac{dt}{dx}$ 知，q 与平壁的材料即物性有关

4.略

5.解：

$$\begin{cases} \dfrac{d}{dr}(r^2\dfrac{dt}{dr})=0 \\ r=r_1, t=t_{w1}(\text{设}t_{w1}>t_{w2}) \\ r=r_2, t=t_{w2} \end{cases}$$

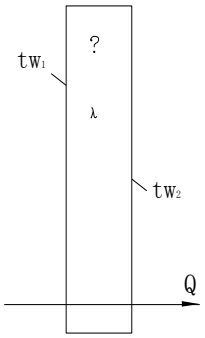


有：
$$Q=\frac{4\pi\lambda}{\frac{1}{r_1}-\frac{1}{r_2}}(t_{w1}-t_{w2})$$

$$R_{\lambda F}=\frac{r_2-r_1}{4\pi\lambda r_2 r_1}$$

6.略

7.已知： $l=4m, h=3m, \delta=0.25$



$t_{w1}=15℃$， $t_{w2}=-5℃$， $\lambda=0.7W/(m\cdot k)$

求：Q

解： $\because l,h \gg \delta$，可认为该墙为无限大平壁

$$\therefore Q=\lambda F\frac{\Delta t}{\delta}=0.7\times(4\times3)\times\frac{15-(-5)}{0.25}=672W$$

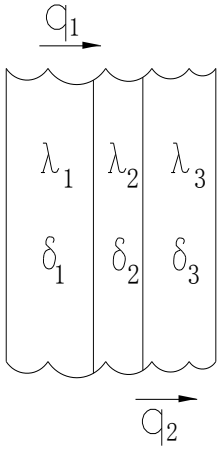
8.已知： $F=20m^2, \delta=0.14m, t_{w2}=-15℃$， $\lambda=1.28W/(m\cdot k), Q=5.5\times10^3W$

求： t_{w1}

解： 由 $Q=\lambda F\dfrac{\Delta t}{\delta}$ 得一无限平壁的稳态导热

$$t_{w1}=t_{w2}+\frac{Q}{\lambda F}\delta=-15+\frac{5.5\times10^3}{20\times1.28}\times0.14=15℃$$

9. 已知：$\delta_1=240mm, \delta_2=20mm$，



$\lambda_1=0.7W/(m\cdot k), \lambda_2=0.58W/(m\cdot k)$

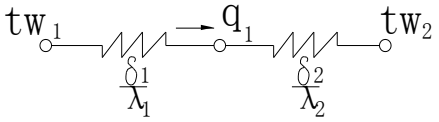
$\lambda_3=0.07W/(m\cdot k), q=150W/m^2$

求： δ_3

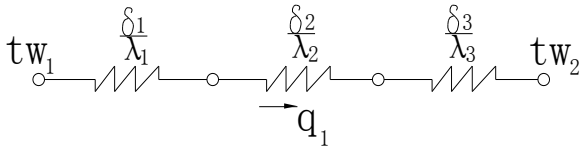
解： 设两种情况下的内外面墙壁温度t_{w1}和t_{w2}保持不变，

且$t_{w1}>t_{w2}$

由题意知：$q_1=\dfrac{t_{w1}-t_{w2}}{\dfrac{\delta_1}{\lambda_1}+\dfrac{\delta_2}{\lambda_2}}$



$$q_2=\frac{t_{w1}-t_{w2}}{\dfrac{\delta_1}{\lambda_1}+\dfrac{\delta_2}{\lambda_2}+\dfrac{\delta_3}{\lambda_3}}$$



再由：　$q_2=0.2q_1$，有

$$\frac{t_{w1}-t_{w2}}{\dfrac{\delta_1}{\lambda_1}+\dfrac{\delta_2}{\lambda_2}+\dfrac{\delta_3}{\lambda_3}}=0.2\frac{t_{w1}-t_{w2}}{\dfrac{\delta_1}{\lambda_1}+\dfrac{\delta_2}{\lambda_2}}$$

得：　$\delta_3=4\lambda_3(\dfrac{\delta_1}{\lambda_1}+\dfrac{\delta_2}{\lambda_2})=4\times0.06\times(\dfrac{240}{0.7}+\dfrac{20}{0.58})=90.6mm$

10.已知：$t_{w1}=450℃$，$\lambda=0.094+0.000125t, t_{w2}=50℃$，$q\le340W/m^2$

求：δ

解：　$q=\lambda m\dfrac{\Delta t}{\delta}, \lambda m=0.094+1.25\times10^{-4}\times\dfrac{t_{w1}+t_{w2}}{2}$

$$\delta=\lambda m\frac{\Delta t}{q}=[0.094+1.25\times10^{-4}\times\frac{t_{w1}+t_{w2}}{2}]\cdot\frac{t_{w1}-t_{w}}{q}$$

$$=[0.094+1.25\times10^{-4}\times\frac{450+50}{2}]\times\frac{450-50}{340}=0.150m$$

即有　$q\le340W/m^2$时有$\delta\ge0.147m$

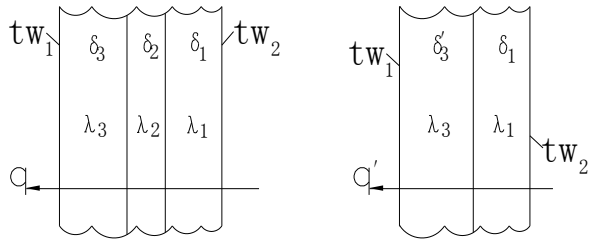
11.已知：$\delta_1=120mm, \lambda_1=0.8W/(m\cdot k)$，$\delta_2=50mm, \lambda_2=0.12W/(m\cdot k)$

$\delta_3=250mm, \lambda_3=0.6W/(m\cdot k)$

求：$\delta_3^{'}=?$

解：　$q=\dfrac{t_{w2}-t_{w1}}{\dfrac{\delta_1}{\lambda_1}+\dfrac{\delta_2}{\lambda_2}+\dfrac{\delta_3}{\lambda_3}}, q^{'}=\dfrac{t_{w2}-t_{w1}}{\dfrac{\delta_1}{\lambda_1}+\dfrac{\delta_3^{'}}{\lambda_3}}$

由题意知：$q=q^{'}$



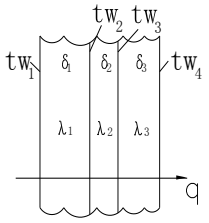
即有：
$$\frac{t_{w2}-t_{w1}}{\frac{\delta_1}{\lambda_1}+\frac{\delta_2}{\lambda_2}+\frac{\delta_3}{\lambda_3}}=\frac{t_{w2}-t_{w}}{\frac{\delta_1}{\lambda_1}+\frac{\delta_3'}{\lambda_3}}$$

$$\delta_3'=\delta_3+\delta_2\frac{\lambda_3}{\lambda_2}$$

$$=250+50\times\frac{0.6}{0.12}=500mm$$

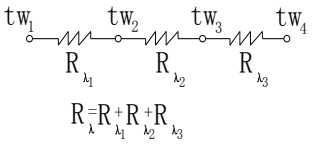
12.已知：$t_{w1}=600℃$，$t_{w2}=480℃$，$t_{w3}=200℃$，$t_{w4}=60℃$

求：$\frac{R_{\lambda1}}{R_\lambda},\frac{R_{\lambda2}}{R_\lambda},\frac{R_{\lambda3}}{R_\lambda}$



解：由题意知其为多层平壁的稳态导热

故有：$q=\frac{t_{w1}-t_{w4}}{R_\lambda}=\frac{t_{w1}-t_{w2}}{R_{\lambda1}}=\frac{t_{w2}-t_{w3}}{R_{\lambda2}}=\frac{t_{w3}-t_{w4}}{R_{\lambda3}}$



$\therefore$
$$\frac{R_{\lambda1}}{R_\lambda}=\frac{t_{w1}-t_{w2}}{t_{w1}-t_{w4}}=\frac{600-480}{600-60}=0.22$$

$$\frac{R_{\lambda2}}{R_\lambda}=\frac{t_{w2}-t_{w3}}{t_{w1}-t_{w4}}=\frac{480-200}{600-60}=0.52$$

$$\frac{R_{\lambda3}}{R_\lambda}=\frac{t_{w3}-t_{w4}}{t_{w1}-t_{w4}}=\frac{200-60}{600-60}=0.26$$

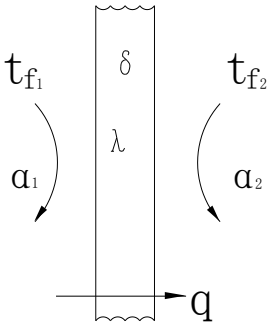
13.略

14.已知：**1**）$\delta_1=2mm,\lambda_1=40W/(m\cdot k),\delta_0=3mm,t_{f1}=250℃$，$t_f=60℃$

$$\lambda_0=\lambda_1,h_1=75W/(m^2\cdot k),h_2=50W/(m^2\cdot k)$$

2）$\delta_2=3mm,\lambda_2=320W/(m\cdot k)$

3）$\delta_3=\delta_0,\lambda_3=\lambda_{0,}h_2'=70W/(m^2\cdot k)$



求：$\Delta q_1,\Delta q_2,\Delta q_3,k_1,k_2,$

解：未变前的

$$q_0=\frac{t_{f1}-t_{f2}}{\frac{1}{h_1}+\frac{\delta_0}{\lambda_0}+\frac{1}{h_2}}=\frac{250-60}{\frac{1}{75}+\frac{3\times10^{-3}}{40}+\frac{1}{50}}=5687.2W/m^2$$

1）$k_1=\dfrac{1}{\dfrac{1}{h_1}+\dfrac{\delta_1}{\lambda_1}+\dfrac{1}{h_2}}=\dfrac{1}{\dfrac{1}{75}+\dfrac{2\times10^{-3}}{40}+\dfrac{1}{50}}=29.96W/(m^2\cdot k)$

$q_1=k_1\Delta t=29.96\times(250-60)=5692.4W/m^2$

$\Delta q_1=q_1-q_0=5692.4-5687=2W/m^2$

2）$k_2=\dfrac{1}{\dfrac{1}{h_1}+\dfrac{\delta_2}{\lambda_2}+\dfrac{1}{h_2}}=\dfrac{1}{\dfrac{1}{75}+\dfrac{3\times10^{-3}}{320}+\dfrac{1}{50}}=29.99W/(m^2\cdot k)$

$q_2=k_2\Delta t=29.99\times(250-60)=5698.4W/m^2$

$\Delta q_2=q_2-q_0=5698.4-5687=11W/m^2$

3) $k_3=\dfrac{1}{\dfrac{1}{h_1}+\dfrac{\delta_0}{\lambda_0}+\dfrac{1}{h_2'}}=\dfrac{1}{\dfrac{1}{75}+\dfrac{3\times10^{-3}}{40}+\dfrac{1}{70}}=36.11W/(m^2\cdot k)$

$q_3=k_3\Delta t=36.11\times(250-60)=6860.7W/m^2$

$\Delta q_3=q_3-q_0=6860.7-5687=1173W/m^2$

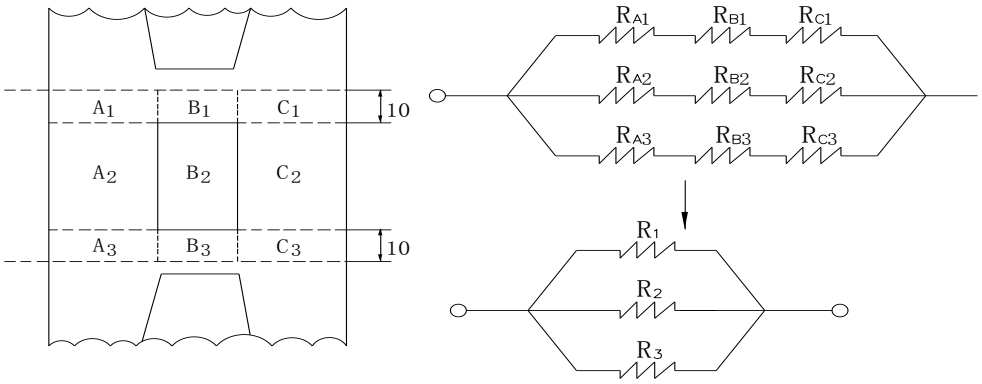
$\therefore \Delta q_3>\Delta q_2>\Delta q_1$，第三种方案的强化换热效果最好

15.已知：$\delta_A=\delta_C=35mm,\delta_B=130mm$，其余尺寸如下图所示，

$\lambda_A=\lambda_C=1.53W/(m\cdot k)\quad\lambda_B=0.742W/(m\cdot k)$

求：R_λ

解：该空斗墙由对称性可取虚线部分，成为三个并联的部分



$R_1 = R_{A1} + R_{B1} + R_{C1} = R_3 \quad R_2 = R_{A2} + R_{B2} +$

$$\therefore R_1 = \frac{\delta_{A1}}{\lambda_{A1}} + \frac{\delta_{B1}}{\lambda_{B1}} + \frac{\delta_{C1}}{\lambda_{C1}} = 2 \times \frac{35 \times 10^{-3}}{1.53} + \frac{130 \times 10^{-3}}{1.53} = R_3 = 0.1307(m^2 \cdot k)/W$$

$$R_2 = \frac{\delta_{A2}}{\lambda_{A2}} + \frac{\delta_{B2}}{\lambda_{B2}} + \frac{\delta_{C3}}{\lambda_{C3}} = 2 \times \frac{35 \times 10^{-3}}{1.53} + \frac{130 \times 10^{-3}}{0.742} = 0.221(m^2 \cdot k)/W$$

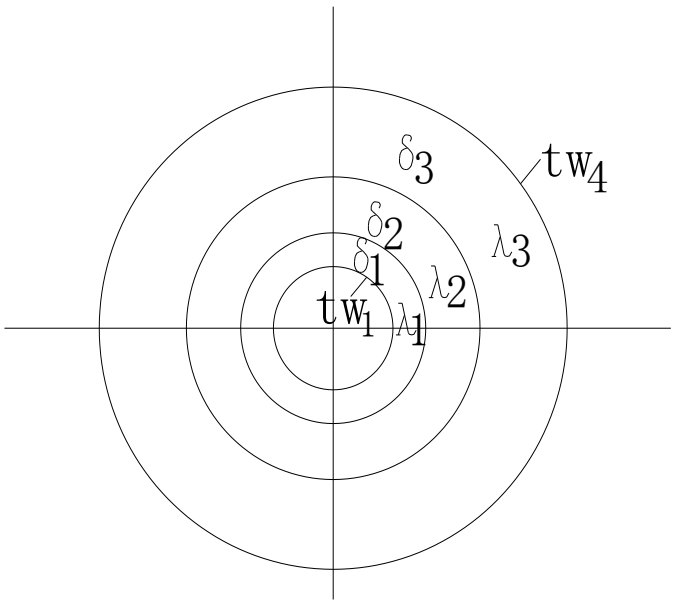
$$\therefore R_\lambda = \frac{1}{2 \times \frac{1}{R_1} + \frac{1}{R_2}} = \frac{1}{2 \times \frac{1}{0.1307} + \frac{1}{0.221}} = 5.04 \times 10^{-2}(m^2 \cdot k)/W$$

16.已知：$d_1 = 160mm, d_2 = 170mm, \lambda_1 = 58W/(m \cdot k)$，$\delta_2 = 30mm, \lambda_2 = 0.093W/(m \cdot k)$

$\delta_3 = 40mm, \lambda_3 = 0.1W/(m \cdot k)$，℃，$t_{w4} = 50$℃

求：1）$R_{\lambda 1}, R_{\lambda 2}, R_{\lambda 3}$； **2)** q_l： **3)** t_{w2}, t_{w3}.

解：



1） $R_{\lambda 1}=\frac{1}{2\pi\lambda_1}\ln\frac{d_2}{d_1}=\frac{1}{2\pi\times 58}\ln\frac{170}{160}=1.664\times10^{-4}(m\cdot k)/W$

$R_{\lambda 2}=\frac{1}{2\pi\lambda_2}\ln\frac{d_2+2\delta_2}{d_2}=\frac{1}{2\pi\times 0.093}\ln\frac{170+60}{170}=0.517(m\cdot k)/W$

$R_{\lambda 3}=\frac{1}{2\pi\lambda_3}\ln\frac{d_2+2\delta_2+2\delta_3}{d_2+2\delta_2}=\frac{1}{2\pi\times 0.17}\ln\frac{170+60+80}{170+60}=0.279(m\cdot k)/W$

$\therefore R_{\lambda 1}$ □ $R_{\lambda 3}<R_{\lambda}$

2) $q_l=\frac{\Delta t}{\sum R_{\lambda i}}=\frac{\Delta t}{R_{\lambda 2}+R_{\lambda 3}}=\frac{300-50}{0.517+0.279}=314.1W/m$

3）由 $q_l=\frac{t_{w1}-t_{w2}}{R_{\lambda 1}}$ 得

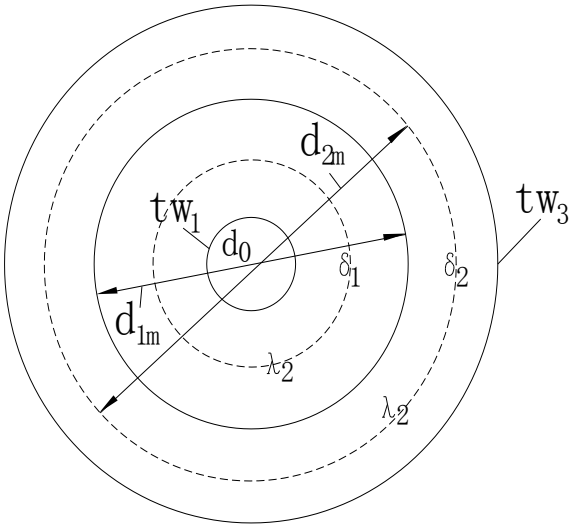
$t_{w2}=t_{w1}-q_l R_{\lambda 1}=300-314.1\times 1.664\times10^{-4}=300$ ℃

同理：

$t_{w3}=t_{w4}+q_l R_{\lambda 3}=50+314.1\times 0.279=137$ ℃

17. 已知：$\delta_1=\delta_2,\lambda_2=\frac{1}{2}\lambda_1,d_{m2}=2d_{m1}$

求：$\frac{q_l}{q_l'}$



解：忽略管壁热阻

$R_{\lambda}=\frac{1}{2\pi\lambda_1}\ln\frac{d_0+2\delta_1}{d_0}+\frac{1}{2\pi\lambda_2}\ln\frac{d_0+2\delta_1+2\delta_2}{d_0+2\delta_1}$

$R_{\lambda}'=\frac{1}{2\pi\lambda_2}\ln\frac{d_0+2\delta_1}{d_0}+\frac{1}{2\pi\lambda_1}\ln\frac{d_0+2\delta_1+2\delta_2}{d_0+2\delta_1}$

$q_l=\frac{\Delta t}{R_{\lambda}},q_l'=\frac{\Delta t}{R_{\lambda}'}$ （管内外壁温t_{w1},t_{w3}不变）

$\therefore\frac{q_l}{q_l'}=\frac{R_{\lambda}'}{R_{\lambda}}=\frac{\frac{1}{2\pi\lambda_2}\ln\frac{d_0+2\delta_1}{d_0}+\frac{1}{2\pi\lambda_1}\ln\frac{d_0+2\delta_1+2\delta_2}{d_0+2\delta_1}}{\frac{1}{2\pi\lambda_1}\ln\frac{d_0+2\delta_1}{d_0}+\frac{1}{2\pi\lambda_2}\ln\frac{d_0+2\delta_1+2\delta_2}{d_0+2\delta_1}}$

$$= \frac{\ln\frac{d_0+2\delta_1}{d_0}+\frac{1}{2}\ln\frac{d_0+4\delta}{d_0+2\delta_1}}{\frac{1}{2}\ln\frac{d_0+2\delta_1}{d_0}+\ln\frac{d_0+4\delta}{d_0+2\delta_1}}$$

由题意知： $d_{1m}=\frac{1}{2}[d_0+(d_0+2\delta_1)]=d_0+\delta_1$

$d_{2m}=\frac{1}{2}[d_{1m}+(d_{1m}+2\delta_2)]=d_0+\delta_2$

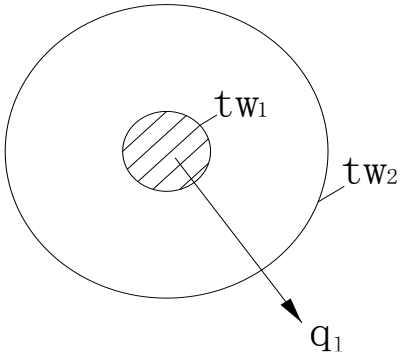
即： $d_{2m}=2d_{1m}\Rightarrow d_0+3\delta_2=2(d_0+\delta_1)\Rightarrow d_0=\delta_2$ （代入上式）

$$\therefore \frac{q_l}{q_l'}=\frac{R_\lambda'}{R_\lambda}=\frac{\ln 3+\frac{1}{2}\ln\frac{5}{3}}{\frac{1}{2}\ln 3+\ln\frac{5}{3}}=1.277$$

即： $q_l'=0.783q_l$

$\Delta=\frac{q_l'-q_l}{q_l}=21.7\%$ 即热损失比原来减小 **21.7%**。

18.已知： $d=1mm,\ R_l=2.22\times10^{-3}\Omega/m,\ \lambda=0.15W/(m\cdot k)$



$t_{w1\max}=65℃$， $t_{w2}=40℃$， $\delta=0.5mm,$

求： $I_{\max}$

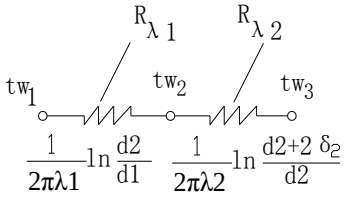
解： $q_l=I_{\max}^2R_l=\frac{t_{w1\max}-t_{w2}}{\frac{1}{2\pi\lambda}\ln\frac{d+2\delta}{d}}$

$$\therefore I_{\max}=\left(\frac{t_{w1\max}-t_{w2}}{\frac{R_l}{2\pi\lambda}\ln\frac{d+2\delta}{d}}\right)^{\frac{1}{2}}=\left(\frac{65-40}{\frac{2.22\times10^{-3}}{2\pi\times0.15}\ln\frac{1+1\times2}{1}}\right)^{\frac{1}{2}}=123.7(A)$$

19.已知： $d_1=85mm, d_2=100mm, \lambda_1=40W/(m\cdot k), t_{w1}=180℃$

$\lambda_2=0.053W/(m\cdot k), t_{w3}\le40℃$， $q_l=52.3W/m$

求： δ_2



解： $q_l=\frac{\Delta t}{R_{\lambda1}+R_{\lambda2}}=\frac{t_{w1}-t_{w3}}{\frac{1}{2\pi\lambda_1}\ln\frac{d_2}{d_1}+\frac{1}{2\pi\lambda_2}\ln\frac{d_2+2\delta_2}{d_2}}$

整理得：

$$\delta_2 = \frac{d_2}{2}(e^{2\pi\lambda_2(\frac{\Delta t}{q_l}-\frac{1}{2\pi\lambda_1}\ln\frac{d_2}{d_1})}-1) = \frac{100}{2}\times(e^{2\pi\times0.053(\frac{180-40}{52.3}-\frac{1}{2\pi\times40}\ln\frac{100}{85})}-1) = 72mm$$

或：$R_{\lambda 2}$ 远大于 $R_{\lambda 1}$，故有 $q_l = \frac{\Delta t}{R_{\lambda 2}} = \frac{t_{w1}-t_{w3}}{\frac{1}{2\pi\lambda_2}\ln\frac{d_2+2\delta_2}{d_2}}$

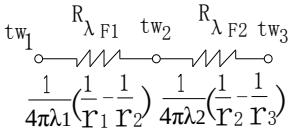
$$\Rightarrow \quad \delta_2 = \frac{d_2}{2}(e^{\frac{2\pi\lambda_2\Delta t}{q_l}}-1) = 72mm$$

20.已知：$d_1 = 0.35mm, \delta_1 = 3mm, \delta_2 = 30mm, r = 199.6kJ/kg, t_{w1} = (-273.15+77.4)℃$

$t_{w3} = 25℃$，$\lambda_2 = 0.03W/(m\cdot k), \lambda_1 = 16.3W/(m\cdot k), \tau = 1h$

求：m

解：$Q = \frac{t_{w3}-t_{w1}}{R_{\lambda F_1}+R_{\lambda F_2}}$



$$= \frac{t_{w3}-t_{w1}}{\frac{1}{4\pi\lambda_1}(\frac{1}{\frac{1}{2}d_1}-\frac{1}{\frac{1}{2}(d_1+2\delta_1)})+\frac{1}{4\pi\lambda_2}(\frac{1}{\frac{1}{2}(d_1+2\delta_1)}-\frac{1}{\frac{1}{2}(d_1+2\delta_1+2\delta_2)})}$$

$$= \frac{2\pi(25+273.15-77.4)}{\frac{1}{16.3}\times(\frac{1}{0.35}-\frac{1}{0.356})+\frac{1}{0.03}\times(\frac{1}{0.356}-\frac{1}{0.35}}$$

$$= 102.7W$$

或：

$R_{\lambda F_1}$ 远大于 $R_{\lambda F_2}$，故有：

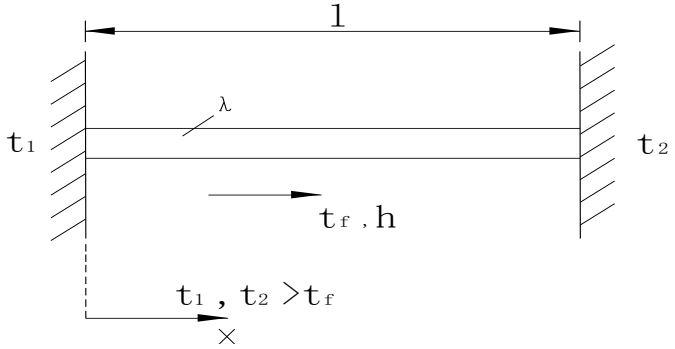
$$Q = \frac{t_{w3}-t_{w1}}{\frac{1}{4\pi\lambda_2}(\frac{1}{r_2}-\frac{1}{r_3})} = \frac{2\pi(25+273.15-77.4)\times0.03}{(\frac{1}{0.356}-\frac{1}{0.416})} = 102.7W$$

$$m = \frac{Q\tau}{r} = \frac{102.7\times3.6}{199.6} = 1.85kg/$$

21.略　　**22.**略

23. 解：

$$\begin{cases} \frac{d^2\theta}{dx^2}-m^2\theta=0, \theta = t-t_f \\ x=0, \theta=\theta_1=t_1-t_f \\ x=l, \theta=\theta_2=t_2-t_f \end{cases}$$



解微分方程可得其通解： $\theta = c_1 e^{mx} + c_2 e^{-mx}$ 由此得温度分布（略）

24.已知： $l = 25mm,\ \delta = 3mm,\ \lambda = 140W/(m\cdot k), h = 75W/(m^2\cdot k), t_0 = 80℃$

$t_f = 30℃$， $q_{x=l} = 0$

求： θ, q_l

解： $ml = \sqrt{\frac{hU}{\lambda A}}l = \sqrt{\frac{h2L}{\lambda\delta L}}l = \sqrt{\frac{2}{\lambda\delta}h}l = 0.025\sqrt{\frac{2\times 75}{140\times 3\times 10^{-3}}} = 0.4$

$m = 18.$

$\theta = \theta_0 \frac{ch[m(l-x)]}{ch(ml)} = (80-30)\frac{ch[0.4725(1-x)]}{ch(0.4725)}$

$= 44.9ch[0.4725(1-x)]$

$\therefore t = 30 + 44.9ch[0.4725(1-x)]$

$q_l = \frac{Q}{L} = \frac{hU}{mL}\theta_0 th(ml) = \frac{2h}{m}\theta_0 th(ml)$

$= \frac{2\times 75}{18.9}\times(80-30)th(0.4725) = 177th(0.4725)W/m$

25.已知： $\delta = 15mm, l = 20mm, \lambda = 48.5W/(m\cdot k), t_l = 84℃$， $t_0 = 40℃$

$h = 20W/(m^2\cdot k)$

求： Δt

解： $ml = \sqrt{\frac{hU}{\lambda A}}l = \sqrt{\frac{h\pi d}{\lambda\pi d\delta}}l = \sqrt{\frac{h}{\lambda\delta}}l = 0.12\times\sqrt{\frac{20}{48.5\times 1.5\times 10^{-3}}} = 2$

$\theta_l = \frac{\theta_0}{ch(ml)} \Rightarrow \frac{t_0 - t_f}{t_l - t_f} = ch(ml)$

$\therefore t_f = \frac{t_l ch(ml) - t_0}{ch(ml)-1} = \frac{84ch(2)-40}{ch(2)-1} = 99.93℃$

$ch(2) = 3.76$

$\Delta t = \frac{t_f - t_l}{t_f}\times 100\% = \frac{99.93-84}{99.93}\times 100\% = 15.9\%$

26.已知： $\delta = 0.8mm, l = 160mm, t_0 = 60℃$， $\lambda = 16.3W/(m\cdot k)$，其他条件同 **25** 题

求：Δt

解：$ml=\sqrt{\frac{h}{\lambda\delta}}l=160\times\sqrt{\frac{20}{16.3\times0.8\times10^{-3}}}=6.27$

$t_f=\frac{t_l ch(ml)-t_0}{ch(ml)-1}=\frac{84ch(6.27)-60}{ch(6.27)-1}=84.09℃$

$ch(6.27)\approx 264$

$\Delta t=\frac{t_f-t_l}{t_f}\times100\%=\frac{84.09-84}{84.09}\times100\%=0.1\%$

27.已知：$\delta=3mm,l=16mm$ （1）$\lambda=140 W/(m\cdot k)$，$h=80 W/m^2\cdot$（

（2）$\lambda=40 W/(m\cdot k)$，$h=125 W/m^2\cdot$（

求：η_f

解：（**1**）$ml=\sqrt{\frac{hU}{\lambda A}}l=\sqrt{\frac{h2L}{\lambda L\delta}}l=\sqrt{\frac{2h}{\lambda\delta}}l=16\times10^{-3}\times\sqrt{\frac{2\times80}{140\times3\times10^{-3}}}=0.312$

$\eta_f=\frac{th(ml)}{ml}=\frac{th(0.312)}{0.312}=0.9$

（**2**）$ml=\sqrt{\frac{hU}{\lambda A}}l=\sqrt{\frac{h2L}{\lambda L\delta}}l=\sqrt{\frac{2h}{\lambda\delta}}l=16\times10^{-3}\times\sqrt{\frac{2\times125}{40\times3\times10^{-3}}}=0.73$

$\eta_f=\frac{th(ml)}{ml}=\frac{th(0.73)}{0.73}=0.85$

28.已知：$d_1=77mm,d_2=140mm,\delta=4mm,P=25mm,\lambda=50W/(m\cdot k)$

$h=60W/(m^2\cdot k),t_0=30℃$，$t_f=75℃$

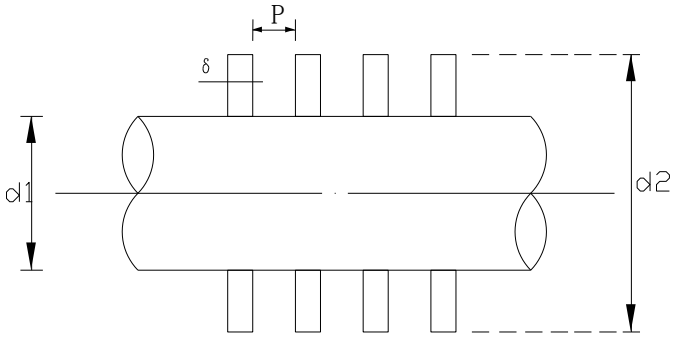
求：q_l

解：$l=\frac{1}{2}(d_2-d_1)=31.5$

$l_c=l+\frac{\delta}{2}=33.$

$r_{2c}=r_1+l_c=72$

$f=\delta(r_{2c}-r_1)=4\times10^{-3}\times(72-38.5)\times10^{-3}=1.34\times10^{-4}m^2$



$$l_c^{\frac{3}{2}}\left(\frac{2h}{\lambda f}\right)^{\frac{1}{2}}=(33.5\times10^{-3})^{\frac{3}{2}}\left(\frac{2\times60}{50\times1.3\times4\times10^{-4}}\right)^{\frac{1}{2}}=0.$$

$$\frac{r_{2c}}{r_1}=\frac{72}{33.5}=2.15$$

查图得：$\eta_f=0.7$

每片肋片的散热量为Q_1

$$Q_1=\eta_f Q_0=\eta_f hF(t_0-t_f)$$

$$=2\pi(r_{2c}^2-r_1^2)\eta_f h(t_0-t_f)$$

$$=2\pi(72^2-33.5^2)\times10^{-6}\times0.7\times60\times(320-W$$

每米肋片管的散热量为：

$$q_l=nQ_1+(n-1)Q_2 \qquad n=\frac{1000}{25}+1=41\text{片/米}$$

$$=41\times266+7\times40\times1.48W$$

Q_2为两肋片间的表面的散热量

$$Q_2=\pi d_1 P(t_0-t_f)$$

$$=\pi\times77\times10^{-3}\times25\times10^{-3}\times(320-75)=1.48W$$

29.略

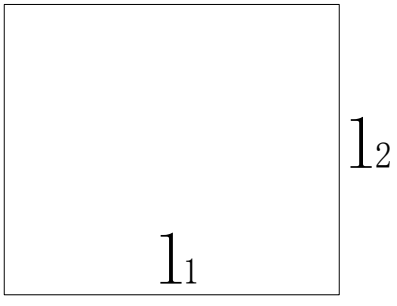
30.已知：$l_1\times l_2=3\times2.2m^2,\delta=0.3m,\lambda=0.56W/(m\cdot k),t_{w1}=0℃,\ t_{w2}=30℃$

求：q_l

解：

$$S_1=\frac{A_1}{\delta}=\frac{l_1\times L}{\delta}=\frac{3\times L}{0.3}=10L$$

$$S_2=\frac{A_2}{\delta}=\frac{l_2\times L}{\delta}=\frac{2.2\times L}{0.3}=7.33L$$



$S_3 = 0.54$　　$l_1, l_2 > \frac{1}{5}\delta$

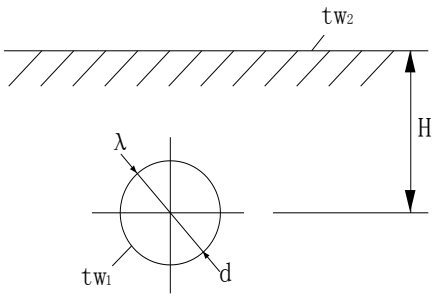
$q_l = \frac{Q}{L} = \frac{(2S_1 + 2S_2 + 4S_3)\lambda\Delta t}{L}$，　　$\Delta t = t_{w1} - t_{w2}$

$= (2\times 10 + 2\times 7.3 + 4\times 0.54)\times 0.56$

$= 618.6\ W/m$

31.已知：$d = 165mm, t_{w1} = 90℃$，$H = 1.5m, \lambda = 1.05W/(m\cdot k), t_{w2} = 6℃$

$h = 20W/(m^2\cdot k)$



求：q_l

解：　$l > r, H > 3r$

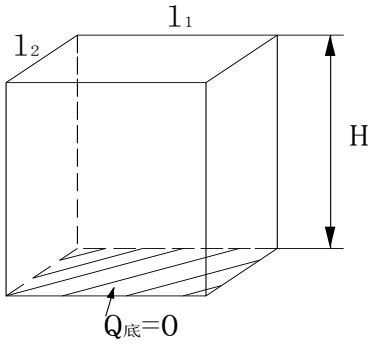
∴　$s = \frac{2\pi l}{\ln\left(\frac{2H}{r}\right)}$

$q_l = \frac{Q}{l} = \frac{s\lambda\Delta t}{l} = \frac{2\pi\lambda}{\ln\left(\frac{2H}{r}\right)}\Delta t$

$= \frac{2\pi\times 1.05}{\ln\left(\frac{2\times 1.5}{0.165/2}\right)}\times(90-6) = 154.2\ W/m$

32.已知：$l_1\times l_2 = 0.52\times 0.52 m^2, H = 0.42m, \lambda = 0.023W/(m\cdot k), t_{w1} = 30℃$

$t_{w2} = -14℃$，　$Q = 34W$



求：δ

解：　$S_1 = \frac{l_1\times l_2}{\delta}, S_2 = \frac{l_1\times H}{\delta}, l_1 = l$

$S_3 = 0.54H$，$S_4 = 0.$

$Q = (S_1 + 4S_2 + 4S_3 + 4S_4)\lambda$

$\delta = \frac{l_1 l_2 + 4l_1 H}{\frac{Q}{\lambda\Delta t} - 4S_3 - 4S_4} = \frac{0.52\times 0.52 + 4\times 0.52\times 0.42}{\frac{34}{0.023\times(30+14)} - 4\times 0.54\times 0.42 - 4\times 0.54\times 0.52}$

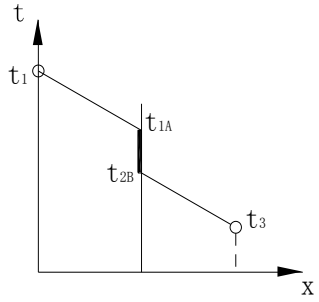
$= 3.62\times 10^{-2} m = 36.2mm$

33.已知：$\delta = 5mm, \Delta = 2.54\mu m, P = 2MPa, \Delta t = 80℃$，$\lambda = 180W/(m\cdot k)$

求：Δt_c

解：由Δ =2.5μm P= 2MP，查表得，$R_c = 0.88\times10^{-4}(m^2\cdot k)/W$

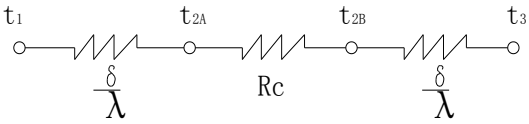
$$Q=\frac{\Delta t}{\frac{\delta}{\lambda}+R_c+\frac{\delta}{\lambda}} \qquad \Delta t = t_3 - t_1$$



再由 $Q=\frac{\Delta t_c}{R_c}$，$\Delta t_c = t_{2A}-t_{2B}$

得

$$\Delta t_c = \frac{R_c}{2\frac{\delta}{\lambda}+R_c}\Delta t = \frac{0.88\times10^{-4}}{2\times\frac{5\times10^{-3}}{180}+0.88\times10^{-4}}\times80 = 4℃$$



第三章 非稳态导热

1.略 2.略 3.略 4.略

5.已知：$d=0.15mm, c_p=420J/(kg\cdot k), \rho=8400kg/m^3, h_1=58W/(m^2\cdot k)$

$h_2=126W/(m^2\cdot k)$

求：τ_{01}, τ_{02}

解：$$\tau_{01}=\frac{\rho c_p V}{h_1 F}=\frac{\rho c_p \frac{4}{3}\pi\left(\frac{d}{2}\right)^3}{h_1 4\pi\left(\frac{d}{2}\right)^2}=\frac{\rho c_p\left(\frac{d}{2}\right)}{3h_1}=\frac{8400\times420\times0.15}{2\times3\times58}\times10^{-3}=1.52(s)$$

同理：$$\tau_{02}=\frac{\rho c_p\left(\frac{d}{2}\right)}{3h_2}=\frac{8400\times420\times0.15}{2\times3\times126}\times10^{-3}=0.7(s)$$

6.略

7.已知：$d=0.5mm, \rho=8930kg/m^3, c_p=400J/(kg\cdot k), t_0=25℃$，$t_f=120℃$

$h=95W/(m^2\cdot k), \frac{\theta}{\theta_0}=1\%, \lambda=22W/(m\cdot k)$（康铜）

求：τ, t

解：由 $$\frac{\theta}{\theta_0}=\frac{t-t_f}{t_0-t_f}=1\%$$

$$\Rightarrow \quad t=t_f+0.01(t_0-t_f)=120+0.01\times(25-120)℃$$

$$\because Biv=\frac{h\frac{V}{F}}{\lambda}=\frac{h\frac{R}{3}}{\lambda}=\frac{95\times0.5\times10^{-3}}{2\times3\times22}=3.6\times10^{-4}<0.1M=\frac{1}{3}\times0.1$$

故满足集总参数法的求解条件，有：$\frac{\theta}{\theta_0}=e^{-Bi_V Fo_V}$

$$\Rightarrow\tau=-\frac{\rho c_p V}{hF}\ln\frac{\theta}{\theta_0}=-\frac{8930\times400\times\frac{1}{3}\times0.5^3\times10^{-3}}{95\times2}\ln(1-0.1)=1s$$

8. 已知：$\delta=3mm,F=1\times1m^2$，$h=39W/(m^2\cdot k),\lambda=48.5W/(m\cdot k)$，$t_0=300$ ℃，

$t_f=20$ ℃，$a=12.7\times10^{-6}m^2/s,t=50$ ℃

求：τ

解：$\because Bi_V=\frac{h\frac{\delta}{2}}{\lambda}=\frac{30\times\frac{3}{2}\times10^{-3}}{48.5}=0.98\times10^{-3}<0.1M=\frac{1}{3}\times0.1$

∴ 满足集总参数法的求解条件，故有：

$$\frac{\theta}{\theta_0}=e^{-\frac{hF}{\rho c_p V}\tau}$$

$$\Rightarrow\tau=-\frac{\rho c_p V}{hF}\ln\frac{\theta}{\theta_0}=-\frac{\lambda V}{haF}\ln\frac{\theta}{\theta_0}$$

$$=\frac{48.5\times\frac{3}{2}\times10^{-3}}{39\times12.7\times10^{-6}}\ln\frac{50-20}{100-20}=32s$$

9.略

10.已知：$t_0=80$ ℃，$d=20mm,t_f=20$ ℃，$u=12m/s,\tau=5\min,t=34$ ℃

$\rho=8954kg/m^3,c_p=383.1J/(kg\cdot k),\lambda=386W/(m\cdot k)$

求：h

解：假设可使用集总参数法，故有：$\frac{\theta}{\theta_0}=e^{-\frac{hF}{\rho c_p V}\tau}$

$$\Rightarrow h=-\frac{\rho c_p V}{\tau F}\ln\frac{\theta}{\theta_0}=-\frac{8954\times383.1\times\frac{1}{2}\times\frac{20}{2}\times10^{-3}}{5\times60}\ln\frac{34-20}{80-20}=83.2W/(m^2\cdot k)$$

由 $Biv=\frac{h\frac{V}{F}}{\lambda}=\frac{h\frac{R}{2}}{\lambda}=\frac{83.2\times20\times10^{-3}}{2\times386}=2.16\times10^{-3}<0.1M=\frac{1}{2}$

∴ 满足集总参数法的计算，上述假设成立。

11.已知：$\delta_A = 2\delta_B, \rho_A = \rho_B, c_{pA} = c_{pB}, t_A = t_B, t_f, h_A = h_B \to \infty, \tau_B = 12\min$

$$\frac{\theta_{mA}}{\theta_0} = 50\% = \frac{\theta_{mB}}{\theta_0}$$

求：τ_A

解：$$\because Bi_A^{-1} = \frac{\lambda_A}{h_A \frac{\delta_A}{2}} = 0 = Bi_A^{-1}, \frac{\theta_{mA}}{\theta_0} = 0.5 = \frac{\theta_{mB}}{\theta_0}$$

查表得：$Fo_A = 0.24 = Fo_B$

即：$$\frac{a_A \tau_A}{\delta_A^2} = \frac{a_B \tau_B}{\delta_B^2} \Rightarrow \tau_A = \tau_B \left(\frac{\delta_A}{\delta_B}\right)^2 = 12 \times 2^2 = 48\min$$

12.已知：$a \times b \times c = 0.5 \times 0.5 \times 0.5 m^3, t_0 = 30℃，\ t_f = 800℃，\ \lambda = 52W/(m \cdot k)$

$$a = 0.063 m^2/h, h = 80W/(m^2 \cdot k), \tau = 30\min$$

求：t_m

解：$$Bi^{-1} = \frac{\lambda}{h\delta} = \frac{52}{80 \times \frac{1}{2} \times 0.5} = 2.6$$

$$Fo = \frac{a\tau}{\delta^2} = \frac{0.063 \times 30 \times 60}{3600 \times 0.25^2} = 0.5$$

对于正六面体有：$\frac{t_f - t_m}{t_f - t_0} = \frac{\theta_m}{\theta_0} = \left(\frac{\theta_m}{\theta_0}\right)_{平板}^3$

由 $Bi^{-1} = 2.6$ $Fo = 0.5$ 查图有：$\left(\frac{\theta_m}{\theta_0}\right)_{平板} = 0.9$

$$\therefore t_m = t_f - \theta_0 \left(\frac{\theta_m}{\theta_0}\right)_{平板}^3 = 800 - (800 - 30)^3 = 0.9℃$$

13. 已知：$\delta = 40mm, a = 5 \times 10^{-7} m^2/s, \lambda = 4W/(m \cdot k), t_0 = 25℃，\ t_f = 1260℃$

$h = 40W/(m^2 \cdot k)$, $\frac{h}{\delta}$, =

缺少

14.已知：$t_f = 40℃$、$t_w = 20℃$、$u_\infty = 0.8 m/s$、$l = 0.45m$、$\mathrm{Re}_c = 5 \times 10^5$、$x_1 = 0.1m$、$x_2 = 0.2m$、$x_3 = 0.3m$ $x_4 = l$

求：α_x　　α

解：$t_m=\frac{1}{2}(t_f+t_w)=30℃$

按$t_m=30℃$　查表得：$\mathrm{Pr}=5.42$、$\lambda=0.618\,{}^{w}\!/_{(m\cdot k)}$、$v=8.05\times10^{-7}\,{}^{m^2}\!/_{s}$

由 $\mathrm{Re}_x=\frac{u_\infty x}{v}$　得　$x_1=0.1m$　　$\mathrm{Re}_x=9.94\times10^4$

$x_2=0.2m$　　$\mathrm{Re}_x=1.99\times10^5$　　$<R_{ec}$ 均为层流

$x_3=0.3m$　　$\mathrm{Re}_x=2.98\times10^5$

$x_4=0.45m$　　$\mathrm{Re}_x=4.47\times10^5$

$\therefore\quad \alpha_x=0.332\frac{\lambda}{\chi}\mathrm{Pr}^{1/3}\mathrm{Re}_x^{1/2}=$　$x_1=0.1m$　1136.3

$x_2=0.2m$　803.9　${}^{w}\!/_{(m^2\cdot k)}$

$x_3=0.3m$　655.8

$x_4=0.45m$　535.5

$\therefore\quad \alpha=2\alpha_x$　　图略

15. 已知：$l=0.3m$、$u_\infty=0.9\,{}^{m}\!/_{s}$、$t_f=25℃$

求：$\delta_{\max}$、$u_l(y)$

解：由$t_f=25℃$查表得 $v=9.055\times10^{-7}\,{}^{m^2}\!/_{s}$

$\therefore\quad \mathrm{Re}_l=\frac{u_\infty l}{v}=\frac{0.3\times0.9}{9.055\times10^{-7}}=2.98\times10^5<\mathrm{Re}_c$

$$\delta_{\max}=\frac{4.64l}{\sqrt{\mathrm{Re}_l}}=\frac{4.64}{\sqrt{2.98\times10^5}}\times0.3=2.55\times10^{-3}m$$

$$u_l(y)=\left[\frac{3}{2}(\frac{y}{\delta_{\max}})-\frac{1}{2}(\frac{y}{\delta_{\max}})^3\right]u_\infty$$

$$=\frac{3}{2}\times0.9\times\frac{1}{2.55\times10^{-3}}y-\frac{1}{2}\times0.9\times(\frac{1}{2.55\times10^{-3}}y)^3$$

$\therefore\ u_l(y)=529.4y-2.71\times10^7 y^3$ m/s 图略

16. 略

17. 略

18. 已知：$u_\infty=10\,m/s$、$t_f=80℃$、$t_w=30℃$、$l=0.8m$、$\mathrm{Re}_c=5\times10^5$、$b=1m$

求：x_c、α、Q

解：$t_m=\frac{1}{2}(t_f+t_w)=55℃$

按 $t_m=55$ ℃ 查表得：$v=1.846\times10^{-5}\,m^2/s$、$\mathrm{Pr}=0.697$、$\lambda=2.865\times10^{-2}\,w/(m\cdot k)$

由 $\mathrm{Re}_c=\frac{u_\infty x_c}{v}\rightarrow x_c=\frac{\mathrm{Re}_c\,v}{u_\infty}=\frac{5\times10^5\times1.864\times10^{-5}}{10}=0.923m$

$\because l<x_c$ $\therefore$全板长均为层流

$$\therefore\alpha=2\alpha_l=0.664\frac{\lambda}{l}\mathrm{Re}_l^{1/2}\,\mathrm{Pr}^{1/3}$$
$$=0.664\times\frac{2.865\times10^{-2}}{0.8}\times\left(\frac{10\times0.8}{1.846\times10^{-5}}\right)^{1/2}\times0.697^{1/3}$$
$$=13.9\,w/(m^2\cdot k)$$

$$Q=\alpha F\Delta t=13.9\times1\times0.8\times(80-30)=556w$$

19. 略

20. 略

21. 已知：$\frac{u}{u_\infty}=\frac{y}{\delta}$ 层流 求：$\frac{\delta}{x}$

解：由动量积分方程有：

$$式左=\rho\frac{d}{dx}\int_0^\delta u(u_\infty-u)dy=\rho\frac{d}{dx}\int_0^\delta u_\infty^{\,2}\cdot\frac{u}{u_\infty}(1-\frac{u}{u_\infty})dy=\rho\frac{d}{dx}\left[u_\infty^{\,2}\int_0^\delta\frac{y}{\delta}(1-\frac{y}{\delta})dy\right]$$
$$=\rho\frac{d}{dx}(\frac{u_\infty^{\,2}\delta}{\sigma})$$

$$式右=\quad u\left.\frac{dU}{dy}\right|_{y=0}=u\frac{U_\infty}{\delta}$$

即：$\rho\frac{d}{dx}(\frac{U_\infty^{\,2}\delta}{\sigma})=u\frac{U_\infty}{\delta}$

即 $U_\infty = cons$ 即有：

$$\frac{d\delta}{dx} = \sigma\frac{\nu}{U_\infty\delta} \Rightarrow \delta d\delta = 6\frac{\nu}{U_\infty}dx$$

两边积分有：$\int_0^\delta \delta d\delta = \int_0^x 6\frac{\nu}{U_\infty}dx \Rightarrow \frac{1}{2}\delta^2 = 6\frac{\nu_x}{u_\infty}$

$$\therefore \frac{\delta}{x} = \frac{\sqrt{12}}{\mathrm{Re}_x^{1/2}} \quad ，\ \mathrm{Re}_x = \frac{U_\infty x}{\nu}$$

22. 略

23. 已知：$t = a - by + cy^2, t_w \quad t_f$

求：α_x

解：由边界层特点知 $y=0 \quad t=t_w \quad \left.\frac{d_t^2}{d_x^2}\right|_{y=0} = 0$

$y=\delta \quad t=t_f$

得：$c=0$ 、 $a=t_w$ 、$b = (t_w - t_f)/\delta$

$$\alpha_x = -\frac{\lambda}{t_w - t_f}\cdot\left.\frac{d_t}{d_y}\right|_{y=0} = -\frac{\lambda}{t_w - t_f}(-b+2cy)\big|_{y=0} = \frac{\lambda}{t_w - t_f}\cdot b = \frac{\lambda}{\delta}$$

24. 略

25. 略 $\alpha = \frac{1}{r}\int_0^{r_0}\alpha_r dr$ ， $Q = \alpha F\Delta t$

26. 略 $\alpha = \frac{1}{l}\int_0^{l}\alpha_x dx$ ， $Q = \alpha F\Delta t$

27. 略 $\alpha = -\frac{\lambda}{t_w - t_f}\left.\frac{dt}{dy}\right|_{y=0}$

28. 已知：$F = 3m^2$ 、 $l_1 = 1m$ 、 $t_{w1} = 140$ ℃、 $t_{f_1} = 30$ ℃、 $u_{\infty 1} = 50 m/s$ 、

$Q_1 = 15000w$、$l_2 = 5m$、$t_{w2} = 20$℃、$t_{f_2} = 70$℃、$u_{\infty 2} = 8 m/s$

求：α_2

解：$t_{m1} = \frac{1}{2}(t_{f_1} + t_{w_1}) = \frac{1}{2}(140+30) = 85$℃

$t_{m2} = \frac{1}{2}(t_{f_2} + t_{w_2}) = \frac{1}{2}(20+70) = 45$℃

按 t_{m1} t_{m2} 查表得：

$\mathrm{Pr}_1 = 0.691$、$\lambda_1 = 3.09\times10^{-2}\ w/(m\cdot k)$、$v_1 = 2.16\times10^{-5}\ m^2/s$

$\mathrm{Pr}_2 = 0.6985$、$\lambda_2 = 2.8\times10^{-2}\ w/(m\cdot k)$、$v_2 = 1.746\times10^{-5}\ m^2/s$

有：$\mathrm{Re}_1 = \dfrac{u_{\infty1}l_1}{v_1} = \dfrac{50\times1}{2.16\times10^{-5}} = 2.31\times10^6$

$\mathrm{Re}_2 = \dfrac{u_{\infty2}l_2}{v_2} = \dfrac{8.08\times5}{1.746\times10^{-5}} = 2.31\times10^6$

知：$\mathrm{Re}_1 = \mathrm{Re}_2$，$\mathrm{Pr}_1 \approx \mathrm{Pr}_2$，且几何相似

得：$Nu_1 = Nu_2$

$$\alpha_2 = \frac{\lambda_2}{l_2}Nu_1 = \frac{\lambda_2}{\lambda_1}\cdot\frac{l_1}{l_2}\cdot\alpha_1$$

而：$\alpha_1 = \dfrac{Q_1}{F_1\Delta t_1}$

$$\therefore \alpha_2 = \frac{\lambda_2}{\lambda_1}\cdot\frac{l_1}{l_2}\cdot\frac{Q_1}{F_1\Delta t_1}$$

$$= \frac{3.8\times10^{-2}}{3.09\times10^{2}}\times\frac{1}{5}\times\frac{15000}{3\times(140-30)} = 8.24\ w/(m^2\cdot k)$$

29. 略

30. 已知：$d_a = 16mm$，$d_b = 30mm$，$G_a = 2G_b$，$t_a = t_b$

求：（1）是否相似（2）如何相似

解：（1）$\because t_a = t_b$，且为同类流体

$\therefore v_a = v_b$，$p_a = p_b$

再：$u_m = \dfrac{G}{\rho F} = \dfrac{G}{\rho\cdot\dfrac{\pi}{4}d^2}$

有：$\frac{u_{ma}}{u_{mb}}=\frac{\frac{G_a}{\rho_a\cdot\frac{\pi}{4}d_a^2}}{\frac{G_b}{\rho_b\cdot\frac{\pi}{4}d_b^2}}=2\frac{d_b^2}{d_a^2}$

得：$\frac{\mathrm{Re}_a}{\mathrm{Re}_b}=\frac{\frac{u_{ma}d_a}{\nu_a}}{\frac{u_{mb}d_b}{\nu_b}}=\frac{u_{ma}\cdot d_a}{u_{mb}\cdot d_b}=2\cdot\frac{d_b^2}{d_a^2}\cdot\frac{d_a}{d_b}=2\frac{d_b}{d_a}\neq 1$

即：$\mathrm{Re}_a\neq\mathrm{Re}_b$

知：两者流态不相似

（2）若要相似，需$\mathrm{Re}_a=\mathrm{Re}_b$

即：$\frac{\mathrm{Re}_a}{\mathrm{Re}_b}=1\rightarrow\frac{u_{ma}d_a}{u_{mb}d_b}=1$

而：$\frac{u_{ma}}{u_{mb}}=\frac{G_ad_b^2}{G_bd_a^2}$，带入上式有：$\frac{G_a}{G_b}\cdot\frac{d_b^2}{d_a^2}\cdot\frac{d_a}{d_b}=1$

$\therefore\frac{G_a}{G_b}=\frac{d_a}{d_b}=\frac{16}{30}=\frac{8}{15}$

即：要使两者相似，两者的质量流量之比应为$8/15$

31. 略

32. 略

33. 略

34. 略

第六章 单相流体对流换热及准则关联式

1、（1）、不同。夏季——热面朝下，冬季——冷面朝下（相当于热面朝上）。

（2）、不同。流动情况及物性不同。

（3）、有影响，高度为其定型尺寸。

（4）、在相同流速下，d 大→Re 大→α 大 （$\mathrm{Re}=\frac{ud}{\upsilon}$）

在相同流量下，d 大→Re 小→α 小 （$u=\frac{\bar{V}}{\frac{\pi}{4}d^2}$）

（5）、略

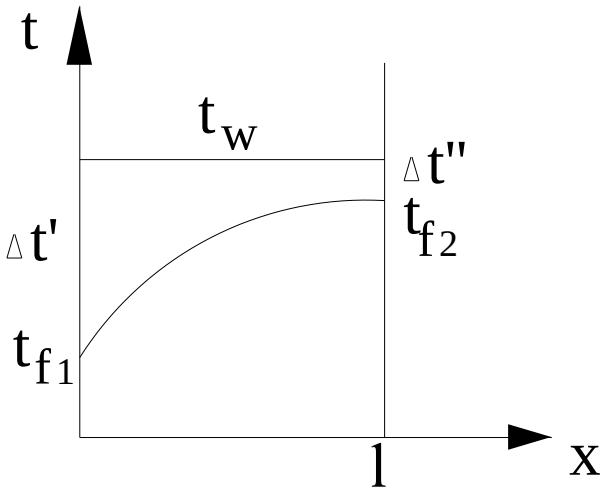
2、略

3、不可以，其不满足边界层类型换热问题所具备的 4 个特征。

4-15、略

16.已知：$t_{f1}=10^{\circ}C, G=0.045kg/s, d=51mm, l=2m, t_w=200^{\circ}C$

求：α, t_{f2}



解：l/d 1(，满足管长条件

设： $t_{f2}=110^{\circ}C$

则：$\Delta t_m=\dfrac{\Delta t'-\Delta t''}{\ln\dfrac{\Delta t'}{\Delta t''}}=\dfrac{190-90}{\ln\dfrac{190}{90}}=133.8^{\circ}C$

$\therefore t_f=t_w-\Delta t_m=200-133.8=66$

按t_f查表有：

$\upsilon_f=19.62\times10^{-6}m^2/s,\ Pr=0.695,\ \lambda_f=2.937\times10^{-2}W/(m\cdot K)$

$Cp_f=1.0075kJ/(kg\cdot K),\ \rho_f=1.041kg/m^3,\ \mu_f=20.41\times10^{-6}N\cdot s/m^2$

$$\mu_m=\frac{G}{\frac{\pi}{4}d^2\rho_f},\ Re_f=\frac{u_m d}{\upsilon}=\frac{G}{\frac{\pi}{4}d\mu_f}=\frac{0.045}{\frac{\pi}{4}\times51\times10^{-3}\times20.41\times10^{-6}}=5\times10^4$$

$$\therefore\alpha=\frac{\lambda_f}{d}\times0.023Re_f^{0.8}Pr_f^{0.4}$$

$$=\frac{2.937\times10^{-2}}{51\times10^{-3}}\times0.023\times(5\times10^4)^{0.8}\times(0.695)^{0.4}=349.7W/(m^2\cdot K)$$

$$Q=GCp_f(t_{f2}-t_{f1})=\alpha\pi dl(t_w-t_f)$$

$$t'_{f2}=t_{f1}+\frac{\alpha\pi dl}{GCp_f}(t_w-t_f)$$

$$=10+\frac{349.7\times51\times10^{-3}\times\pi\times2}{0.045\times1.0075\times10^3}(200-133.8)=104.1^{\circ}C$$

相比较知：$t_{f2}\neq t'_{f2}$，故原假设不合理，重新假设：$t_f=70^{\circ}C$，重复上述步骤，

直至$t_{f2}\approx t'_{f2}$，符合计算精度要求，结果略。($t_f=85C$

17.已知：$d=12.6mm, u_m=1.8m/s, t_w=80^{\circ}C, t_{f1}=28^{\circ}C, t_{f2}=34^{\circ}C, l/d>20$

求：α

解：$\frac{\Delta t'}{\Delta t''}=\frac{80-28}{80-34}=1.13<2$

$\therefore t_f=\frac{1}{2}(t_{f1}+t_{f2})=31^{\circ}C$

按$t_f=31^{\circ}C$查表得：

$\lambda_f=0.62W/(m\cdot K),\upsilon_f=7.9\times10^{-7}m^2/s,\mathrm{Pr}_f=5.31$

$\mathrm{Re}_f=\frac{u_m d}{\upsilon_f}=\frac{1.8\times12.6\times10^{-3}}{7.9\times10^{-7}}=2.87\times10^4>1\times10^4$

∴管内流动处于旺盛紊流

（1）按迪图斯-贝尔特公式计算

$Nu_f=0.023\mathrm{Re}_f^{0.8}\mathrm{Pr}_f^{0.4}$，$t_f$ l ，d，

$\alpha=\frac{\lambda_f}{d}Nu_f=0.023\times(2.87\times10^4)^{0.8}\times(5.31)^{0.4}\times\frac{0.62}{12.6\times10^{-3}}=8130W/(m^2\cdot K)$

（2）按西得-塔特公式计算：

$Nu_f=0.027\mathrm{Re}_f^{0.8}\mathrm{Pr}_f^{1/3}\left(\frac{\mu_f}{\mu_w}\right)^{0.1}$

$\therefore\alpha=\frac{\lambda_f}{d}Nu_f=0.027\times(2.87\times10^4)^{0.8}\times(5.31)^{1/3}\times\frac{0.62}{12.6\times10^{-3}}\times\left(\frac{7.87\times10^{-4}}{3.55\times10^{-4}}\right)^{0.1}$

$=9540W/(m^2\cdot K)$

18.已知：$t_w=250^{\circ}C,t_{f1}=160^{\circ}C,t_{f2}=240^{\circ}C,u_m=1m/s,q=3.84\times10^5W/m^2$

求：d,l

解：由$q=coust$知：$t_f=\frac{1}{2}(t_{f1}+t_{f2})=200^{\circ}C$

按t_f查表得：

$\upsilon_f=0.158\times10^{-6}m^2/s$，$\mu_f=1.364\times10^{-4}N\cdot s/m^2$，P

$\lambda_f=66.8\times10^{-2}W/(m\cdot K)$，$\mu_w=1.049\times10^{-4}N\cdot s/m$

假设管内流动为紊流，$\mathrm{Re}_f=\frac{u_m d}{\upsilon_f}$

而 $q=\alpha(t_w-t_f) \rightarrow \alpha=q/(t_w-t_f)$

另：$Nu_f=0.027\mathrm{Re}_f^{0.8}\mathrm{Pr}_f^{1/3}\left(\frac{\mu_f}{\mu_w}\right)^{0.1} \rightarrow \alpha=0.027\frac{\lambda_f}{d}\mathrm{Re}_f^{0.8}\mathrm{Pr}_f^{1/3}\left(\frac{\mu_f}{\mu_w}\right)^{0.1}$

得：$d=\left[0.027\left(\frac{\mu_m}{\upsilon_f}\right)^{0.8}\mathrm{Pr}_f^{1/3}\left(\frac{\mu_f}{\mu_w}\right)^{0.14}\frac{\lambda_f}{q}(t_w-t_f)\right]^5$

$$=\left[0.027\left(\frac{1}{0.158\times10^{-6}}\right)^{0.8}(0.693)^{1/3}\left(\frac{1.36\times10^{-4}}{1.09\times10^{-4}}\right)^{0.14}\times\frac{0.663}{3.84\times10^5}(160-250)\right]^5$$

$$=0.111m$$

而：$\mathrm{Re}_f=\frac{u_m d_i}{\upsilon_f}=\frac{0.114\times1}{0.158\times10^{-6}}=7.2\times10^5>10^4$ 原假设成立

再由：$Q=\pi dlq=GCp_f(t_{f2}-t_{f1}) \quad G=\frac{\pi}{4}d^2\rho_f u_m$

得：$l=\frac{d_i\rho_f Cp_f(t_{f2}-t_{f1})}{4q}=\frac{0.114\times863\times4.505\times10^3\times(240-160)}{4\times3.84\times10^5}=23.1m$

且有：$l/d=23.1/0.114$，满足 Nu_f 计算关联式的要求。

19.已知：$d=12mm, D=180mm, n=4, t_{f1}=20^\circ C, u_m=1.7m, t_w=90^\circ C$

求：t_{f2}

解：设 $t_{f2}=40^\circ C$ 有：$\frac{\Delta t'}{\Delta t''}=\frac{90-20}{90-40}<2$，即 $t_f=\frac{1}{2}(t_{f1}+t_{f2})=30^\circ C$

按 $t_f=30^\circ C$ 查表有：

$\rho_f=995.7kg/m^3, Cp_f=4.174kJ/(kg\cdot K), \lambda_f=0.618W/(m\cdot K)$

$\upsilon_f=0.805\times10^{-6}m^2/s, \mathrm{Pr}_f=5.42$

得：$\mathrm{Re}_f=\frac{u_m d}{\upsilon_f}=\frac{12\times10^{-3}\times1.7}{0.805\times10^{-6}}=2.53\times10^4>1\times10^4$

$t_w>t_f$，取：$Nu_f=0.023\mathrm{Re}_f^{0.8}\mathrm{Pr}_f^{0.4}\varepsilon_R \quad \varepsilon_R=1+10.3\left(\frac{d}{R}\right)^3$

有：$\alpha=\frac{\lambda_f}{d}Nu_f=0.023\times(2.53\times10^4)^{0.8}\times(5.42)^{0.4}\times\frac{0.628}{0.012}\times\left(1+10.3\times\left(\frac{12}{90}\right)^3\right)$

$=777W/m^2\cdot K$

再由：$Q=\pi dl\alpha(t_w-t_f)=GCp_f(t_{f2}-t_{f1}^{'})$

得：$t_{f2}^{'}=t_{f1}+\frac{\pi dl\alpha}{GCp_f}(t_w-t_f)$

$=t_{f1}+\frac{\pi d\cdot n\pi D\cdot\alpha}{\rho_f u_m\cdot\frac{\pi}{4}d^2Cp_f}(t_w-t_f)=t_{f1}+\frac{4n\pi D\cdot\alpha}{\rho_f Cp_f u_m d}(t_w-t_f)$

$=20+\frac{4\times4\times180\times7774\times10^{-3}\pi(90-30)}{995.7\times4.174\times10^3\times1.7\times12\times10^{-3}}=70^\circ C$

$t_{f2}^{'}\neq t_{f2}$，与假设不符，重新假设t_{f2}的值，重复上述的步骤，直至计算得满足要求的值。结果略！$(t_f=65^\circ C)$

20.略

21.已知：$d=0.16m,l=2.5m,U=5V,I=911.1A,t_{f1}=47^\circ C,u_m=0.5m/s$，$q_w=0$

求：$\alpha,\Delta t=t_w-t_f$

解：$Q=IU=5\times911.1=4555W$

$Q=\alpha F\Delta t=GCp_f(t_{f2}-t_{f1})$

取$Cp_f=4.18\times10^3J/(kg\cdot K),\rho_f=980kg/m^3$

$t_{f2}=t_{f1}+\frac{Q}{GCp_f}=t_{f1}+\frac{Q}{\frac{\pi}{4}d^2\rho_f u_m Cp_f}$

$=47+\frac{4555.5}{\frac{\pi}{4}\times0.16^2\times980\times0.5\times4.18\times10^3}=47.11^\circ C$

可按$t_f=t_{f1}=47^\circ C$查取物性：

$\lambda_f=0.644W/(m\cdot K),\upsilon_f=0.587\times10^{-6}m^2/s,\mathrm{Pr}_f=3.77$

有：$\mathrm{Re}_f=\frac{u_m d}{\upsilon_f}=\frac{0.5\times0.16}{0.587\times10^{-6}}=1.36\times10^5>1\times10^4$

且：$l/d>10$，$t_w>t_f$

取：$Nu_f=0.023Re_f^{0.8}Pr_f^{0.4}$

$$\alpha=\frac{\lambda_f}{d}Nu_f=0.023\times(1.36\times10^5)^{0.8}\times(3.77)^{0.4}\times\frac{0.644}{0.16}=2013W/(m^2\cdot K)$$

$$\Delta t=\frac{Q}{\alpha F}=\frac{Q}{\pi dl\alpha}=\frac{4555.5}{0.16\times2.5\times2013\pi}=1.8^\circ C$$

22.已知：$G=2.5kg/s,t_{f1}=40^\circ C,d=50mm,t_w=85^\circ C,\Delta=0.0002,l=10m$

求：t_{f2},Q

解：设$t_{f2}=70^\circ C,t_f=\frac{1}{2}(t_{f1}+t_{f2})=55^\circ C$

按$t_f=55^\circ C$查表得：

$Cp_f=1.005kJ/(kg\ K)$，$\lambda=2.87\times10^{-2}W/(m\cdot K)$，$\nu=18.46\times10^{-6}m^2/s$

$\Pr_f=0.697,\rho_f=1.077kg/m^3$

$$u_m=\frac{G}{\rho_f\cdot\frac{\pi}{4}d^2},f=\left[2\times\lg\left(\frac{1}{\Delta}\right)+1.74\right]^{-2}$$

由：$St\cdot\Pr^{2/3}=\frac{f}{8}$得$\left(St=\frac{\alpha}{\rho_f Cp_f u_m}\right)$

$$\alpha=\frac{\rho_f Cp_f u_m}{8\Pr^{2/3}\left[1.74+2\lg\left(\frac{1}{\Delta}\right)\right]^2}=\frac{1.077\times1.005\times10^3\times\frac{2.5}{1.077\times\frac{\pi}{4}\times0.05^2}}{8\times(0.697)^{2/3}\times\left[1.74+2\lg\left(\frac{1}{0.0002}\right)\right]^2}$$

$=2431W/(m^2\cdot K)$

再由：$Q=GCp_f(t'_{f2}-t_{f1})=\alpha F(t_w-t_f)$

得：$t'_{f2}=t_{f1}+\frac{\alpha F}{GCp_f}(t_w-t_f)$　　$F=\pi d$

$=40+\frac{2431\times1.0\times0.05\pi}{2.5\times1.005\times10^3}(85-55)=68^\circ C$

$t'_{f2}\neq t_{f2}$，与假设不符，重新假设t_{f2}的值，重复上述的步骤，直至计算得满足要

求的值。结果略！($t_f = 75^\circ C, Q = 3.67\times10^5 W = 367kW$)

23.略

24.已知：$u_m = 1.3m/s, d = 19mm, l = 5.5m, \Delta P = 42mmHg, t_w = 80^\circ C, t_f = 55^\circ C$

求：α

解：按$t_f = 55^\circ C$查表得：

$$\rho_f = 985.6kg/m^3, Cp_f = 4.177kJ/(kg\cdot K), \lambda_f = 65.3\times10^{-2}W/(m\cdot K)$$

$$\upsilon_f = 0.517\times10^{-6}m^2/s, \mathrm{Pr}_f = 3.265$$

知：$$f = \frac{\Delta P}{\frac{l}{d}\cdot\frac{1}{2}\rho_f u_m^{\ 2}} = \frac{42\times133.32}{\frac{5.5}{0.019}\times\frac{1}{2}\times985.6\times1.3^2} = 2.323\times10^{-2}$$

由：$St_f\cdot\mathrm{Pr}_f^{2/3} = \frac{f}{8}$　　($St = \frac{\alpha}{\rho_f Cp_f u_m}$)得：

$$\alpha = \rho_f Cp_f u_m\cdot\frac{f}{8}\cdot\mathrm{Pr}_f^{-2/3}$$

$$= \frac{985.6\times4.177\times10^3\times1.3\times2.323\times10^{-2}}{8\times3.265^{2/3}} = 7061W/(m^2\cdot K)$$

若为光滑量，则有：$\mathrm{Re}_f = \frac{u_m d}{\upsilon_f} = \frac{1.3\times0.019}{0.517\times10^{-6}} = 4.78\times10^4 > 10^4$

知：$Nu_f = 0.023\mathrm{Re}_f^{0.8}\mathrm{Pr}_f^{0.4}$

$$\alpha' = \frac{\lambda_f}{d}\times0.023\mathrm{Re}_f^{0.8}\mathrm{Pr}_f^{0.4} = \frac{0.653}{0.019}\times0.023\times(4.78\times10^4)^{0.8}\times10^{0.4}$$

$$= 7031W/(m^2\cdot K)$$

相比较有：$\alpha \approx \alpha'$

25.已知：$u_m = 1.27m/s, t_f = 38.5^\circ C, t_w = 57.9^\circ C, d = 22mm, l = 2.5m$

求：α

解：按$t_f = 38.5^\circ C$查表得：

$$\lambda_f = 0.0269W/(m\cdot K), \upsilon_f = 16.74\times10^{-6}m^2/s, \mathrm{Pr}_f = 0.7113$$

知：$\mathrm{Re}_f = \frac{u_m d}{\upsilon_f} = \frac{1.27\times22\times10^{-3}}{16.74\times10^{-6}} = 1.67\times10^3$，层流

而：$l/d>60$，则：$Nu_f=0.0214(Re_f^{0.8}-100)Pr_f^{0.4}\left[1+\left(\frac{d}{l}\right)^{2/3}\right]\left(\frac{T_f}{T_w}\right)^{0.4}$

得：$\alpha=0.0214\frac{\lambda_f}{d}(Re_f^{0.8}-100)Pr_f^{0.4}\left[1+\left(\frac{d}{l}\right)^{2/3}\right]\left(\frac{T_f}{T_w}\right)^{0.4}$

$=0.0214\times\frac{0.0269}{0.022}\times\left[(1.67\times10^{4})^{0.8}-100\right]\times(0.71)^{0.4}\left[1+\left(\frac{0.022}{2.5}\right)^{2/3}\right]\left(\frac{330.1}{333.1}\right)^{0.4}$

$=6.2W/(m^2\cdot K)$

26.已知：$u_m=3.5m/s,t_f=58.1^{\circ}C,t_w=90^{\circ}C,$其他同25题

求：α

解：按$t_f=58.1^{\circ}C$查表得：

$\lambda_f=0.02836W/(m\cdot K),\upsilon_f=18.72\times10^{-6}m^2/s,\Pr_f=0.7092$

知：$Re_f=\frac{u_m d}{\upsilon_f}=\frac{3.5\times0.022}{18.72\times10^{-6}}=4.1\times10^{4}$，过渡流

$\alpha=0.0214(Re_f^{0.8}-100)Pr_f^{0.4}\left[1+\left(\frac{d}{l}\right)^{2/3}\right]\left(\frac{T_f}{T_w}\right)^{0.45}=15.78W/(m^2\cdot K)$

27.已知：$d_1=12mm,d_2=16mm,l=400mm,u_m=2.4m/s,t_f=73.1^{\circ}C,t_w=96^{\circ}C$

求：α

解：按$t_f=73.1^{\circ}C$查表得：

$\lambda_f=0.67W/(m\cdot K),\upsilon_f=0.3995\times10^{-6}m^2/s,\Pr_f=2.445$

而：$d_e=\frac{4F}{U}=4\times\frac{\frac{\pi}{4}\left(d_2^{2}-d_1^{2}\right)}{\pi(d_2+d_1)}=d_2-d_1=4mm$

$Re_f=\frac{u_m d_e}{\upsilon_f}=\frac{4\times10^{-3}\times2.4}{0.3995\times10^{-6}}=2.4\times10^{4}>10^{4}$

故可有：$\alpha=\frac{\lambda_f}{d_e}\times0.023Re_f^{0.8}Pr_f^{0.4}$

$=0.023\times\frac{0.67}{0.004}\times(2.4\times10^{4})^{0.8}(2.445)^{0.4}=8897W/m^2K$

28.已知：$d=50mm, G=0.0125kg/s, l=6m, t_{f1}=73.1^{\circ}C, t_{f2}=62^{\circ}C$

求：t_w, α, Q

解：按$t_f=\frac{1}{2}(t_{f1}+t_{f2})=42.75^{\circ}C$查表得：

$\lambda_f=0.0275W/(m\cdot K), \upsilon_f=17.24\times10^{-6}m^2/s, \mathrm{Pr}_f=0.711, Cp_f=1.009kJ/(kg\cdot K)$

$\rho_f=1.175kg/m^3$

有：$\mathrm{Re}_f=\frac{u_m d}{\upsilon_f}, u_m=\frac{G}{\rho_f\cdot\frac{\pi}{4}d^2}$

得：$\mathrm{Re}_f=\frac{\frac{G}{\rho_f\cdot\frac{\pi}{4}d^2}d}{\upsilon_f}=1.65\times10^4>10^4$

故有：$\alpha=0.023\frac{\lambda_f}{d}\mathrm{Re}_f^{0.8}\mathrm{Pr}_f^{0.4}=26.11W/(m^2\cdot K)$

再由：$Q=GCp_f(t_{f2}-t_{f1})=\alpha\cdot\pi dl\cdot(t_w-t_f)$得：

$$t_w=t_f+\frac{GCp_f(t_{f2}-t_{f1})}{\alpha\cdot\pi dl}=62.5^{\circ}C$$

再按：$t_f=t_w-\Delta t_m=t_w-\frac{\Delta t'-\Delta t''}{\ln\frac{\Delta t'}{\Delta t''}}=53.66^{\circ}C$查表得：

$\lambda_f=0.02805W/(m\cdot K), \upsilon_f=18.29\times10^{-6}m^2/s, \mathrm{Pr}_f=0.7097$

$\rho_f=1.0808kg/m^3, Cp_f=1.009kJ/(kg\cdot K)$

计算得（步骤同上）：

$u_m=5.89m/s, \mathrm{Re}_f=1.61\times10^4>10^4$

有：$\alpha=0.027\frac{\lambda_f}{d}\mathrm{Re}_f^{0.8}\mathrm{Pr}_f^{1/3}\left(\frac{T_f}{T_w}\right)^{0.55}=30W/(m^2\cdot K)$

$Q=GCp_f(t_{f2}-t_{f1})=485.6W$

$t_w=t_f+\frac{Q}{\alpha\cdot\pi dl}=70.8^{\circ}C$

29.略　　30.略

31.已知： $u_m = 25.5m/s, d = 35mm, l = 0.5m, Q = 900W, t_f = 25.3^\circ C$

求： t_w

解：按 $t_f = 25.3^\circ C$ **查表得：**

$$\upsilon_f = 15.6\times10^{-6} m^2/s, \lambda_f \approx 0.0261 W/(m\cdot K), \mathrm{Pr}_f = 0.7125$$

$$\mathrm{Re}_f = \frac{u_m d}{\upsilon_f} = \frac{25.5\times0.035}{15.6\times10^{-6}} = 5.72\times10^4$$

查表知： $c = 0.26, n = 0.6$，取 $\varepsilon_t = 1$，$\mathrm{Pr}_w = \mathrm{Pr}_f$，**即：**

$$Nu_f = 0.26\,\mathrm{Re}_f^{0.6}\mathrm{Pr}_f^{0.37}\left(\frac{\mathrm{Pr}_f}{\mathrm{Pr}_w}\right)^{0.25}$$

$$\therefore \alpha = 0.26\times(0.7125)^{0.37}\times(5.72\times10^4)^{0.6}\times\frac{0.0261}{0.035} = 122 W/(m^2\cdot K)$$

再由： $Q = \alpha\cdot\pi dl\cdot(t_w - t_f)$ **得：**

$$t_w = t_f + \frac{Q}{\alpha\cdot\pi dl} = 159.6^\circ C$$

32.已知： $d = 14mm, l = 1.5m, u_\infty = 3m/s, t_f = 55^\circ C, t_w = 95^\circ C$

求： Q

解：按 $t_f = 55^\circ C$ **查表得：**

$$\upsilon_f = 0.517\times10^{-6} m^2/s, \lambda_f = 66.35\times10^{-2} W/(m\cdot K), \mathrm{Pr}_f = 3.26$$

按 $t_w = 95^\circ C$ **查表得：** $\mathrm{Pr}_w = 1.85$

有： $\mathrm{Re}_f = \frac{u_\infty d}{\upsilon_f} = \frac{3\times0.014}{0.517\times10^{-6}} = 8.12\times10^4$

查表得： $c = 0.26, n = 0.6$，**即：** $Nu_f = 0.26\,\mathrm{Re}_f^{0.6}\,\mathrm{Pr}_f^{0.37}\left(\frac{\mathrm{Pr}_f}{\mathrm{Pr}_w}\right)^{0.25}$

$$\alpha = 0.26\frac{\lambda_f}{d}\mathrm{Re}_f^{0.6}\,\mathrm{Pr}_f^{0.37}\left(\frac{\mathrm{Pr}_f}{\mathrm{Pr}_w}\right)^{0.25} = 19.4 kW/(m^2\cdot K)$$

$$Q = \alpha\cdot\pi dl\cdot(t_w - t_f) = 51.2kW$$

33.已知：$d=12mm,u_\infty=14m/s,t_f=30.1^\circ C,t_w=12^\circ C$

求：α

解：按$t_f=55^\circ C$查表得：

$\upsilon_f=16.04\times10^{-6}m^2/s,\lambda_f=0.0264W/(m\cdot K)$

有：$\mathrm{Re}_f=\dfrac{u_\infty d}{\upsilon_f}=\dfrac{14\times0.012}{16.04\times10^{-6}}=1.05\times10^4$

按Re_f查表得：$c=0.26,n=0.6$，即：$Nu_f=0.88c\,\mathrm{Re}_f^n$

$\alpha=0.88\times0.26\times(1.05\times10^4)^{0.6}=130.2W/(m^2\cdot^\circ C)$

34.已知：$P=6,u_{\max}=15.5m/s,t_f=19.4^\circ C,t_w=67.8^\circ C,\dfrac{S_1}{d}=\dfrac{S_2}{d}=1.2,d=19mm$

求：α

解：按$t_f=19.4^\circ C$查表得：

$\upsilon_f=15.06\times10^{-6}m^2/s,\lambda_f=0.02567W/(m\cdot K),\mathrm{Pr}_f=0.713$

$\mathrm{Pr}_w=0.709,\varepsilon_z=0.95$

有：$\mathrm{Re}_f=\dfrac{u_{\max}d}{\upsilon_f}=\dfrac{15.5\times0.019}{15.06\times10^{-6}}=1.96\times10^4$

查表有：$Nu_f=0.027\,\mathrm{Re}_f^{0.63}\,\mathrm{Pr}_f^{0.36}\left(\dfrac{\mathrm{Pr}_f}{\mathrm{Pr}_w}\right)^{0.25}\varepsilon_z$

$\therefore\alpha=0.027\dfrac{\lambda_f}{d}\mathrm{Re}_f^{0.63}\,\mathrm{Pr}_f^{0.36}\left(\dfrac{\mathrm{Pr}_f}{\mathrm{Pr}_w}\right)^{0.25}\varepsilon_z=155.5W/(m^2\cdot K)$

35.已知：$P=5,u_{\max}=4.87m/s,t_f=20.2^\circ C,t_w=25.2^\circ C,\dfrac{S_1}{d}=\dfrac{S_2}{d}=1.25,d=19mm$

求：α

解：按$t_f=20.2^\circ C$查表得：

$\upsilon_f=1.002\times10^{-6}m^2/s,\lambda_f=59.94\times10^{-2}W/(m\cdot K),\mathrm{Pr}_f=6.99,\mathrm{Pr}_w=6.14$

有：$\mathrm{Re}_f=\dfrac{u_{\max}d}{\upsilon_f}=\dfrac{4.87\times0.019}{1.002\times10^{-6}}=9.23\times10^4$

查表有：$\varepsilon_z=0.92$

$$Nu_f = 0.35\,\mathrm{Re}_f^{0.6}\,\mathrm{Pr}_f^{0.36}\left(\frac{\mathrm{Pr}_f}{\mathrm{Pr}_w}\right)^{0.25}\left(\frac{S_1}{S_2}\right)^{0.2}\varepsilon_z$$

$$\therefore \alpha = 0.35\frac{\lambda_f}{d}\,\mathrm{Re}_f^{0.6}\,\mathrm{Pr}_f^{0.36}\left(\frac{\mathrm{Pr}_f}{\mathrm{Pr}_w}\right)^{0.25}\left(\frac{S_1}{S_2}\right)^{0.2}\varepsilon_z = 21.1W/(m^2\cdot K)$$

36.已知：$P=12, d=25mm, S_1=50mm, S_2=45mm, u_{\max}=5m/s, t_f=60^\circ C$

求：$\alpha_{叉}$，$\alpha_{顺}$

解：按$t_f=60^\circ C$查表得：

$$\upsilon_f = 18.9\times10^{-6}m^2/s, \lambda_f = 0.0285W/(m\cdot K)$$

有：$\mathrm{Re}_f = \frac{u_{\max}d}{\upsilon_f} = \frac{5\times0.025}{18.9\times10^{-6}} = 6.61\times10^3$

对于叉排查表得：$\varepsilon_z = 0.98$，另：$\frac{S_1}{S_2} = 0.9$

$$Nu_f = 0.31\mathrm{Re}_f^{0.6}\left(\frac{S_1}{S_2}\right)^{0.2}\varepsilon_z$$

$$\therefore \alpha = 0.31\frac{\lambda_f}{d}\mathrm{Re}_f^{0.6}\left(\frac{S_1}{S_2}\right)^{0.2}\varepsilon_z = 66.4W/(m^2\cdot K)$$

对于顺排有：$\varepsilon_z' = 0.98$

$$\therefore \alpha = 0.24\frac{\lambda_f}{d}\mathrm{Re}_f^{0.6}\,\varepsilon_z' = 68.4W/(m^2\cdot K)$$

37-39.略

40.已知：$d=30mm, t_f=37.1^\circ C, t_w=64.5^\circ C$

求：α

解：按$t_m=\frac{1}{2}(t_f+t_w)=50.8$查表得：

$$\upsilon_m = 0.55\times10^{-6}m^2/s, \mathrm{Pr}_m = 3.495, \lambda_m = 64.89\times10^{-2}W/(m\cdot K), \beta_m = 4.63\times10^{-4}\,1/K$$

故有：$(Gr\,\mathrm{Pr})_m = \frac{g\beta_m\Delta t d^3}{\upsilon_m^2}\,\mathrm{Pr}_m$

$$= \frac{9.81\times(64.5-37.1)\times(0.03)^3}{4.63\times10^{-4}\times(0.55\times10^{-6})^2}\times3.49 = 1.81\times10^8$$

查表得：$c=0.48, n=1/4$

$\therefore Nu_m=0.48(Gr\ Pr)_m^{1/4}$

$$\alpha=0.48\frac{\lambda_m}{d}(Gr\,\mathrm{Pr})_m^{1/4}=0.48\times\frac{0.6489}{0.03}\times(1.81\times10^6)^{1/4}=380.1W/(m^2\cdot K)$$

作为常壁温处理的原因：在水中的自然对换热。

41.已知：$d=50mm, h=0.5m, t_w=90^\circ C, t_f=20^\circ C$

求：α

解：按$t_m=\frac{1}{2}(t_f+t_w)=55^\circ C$查表得：

$$\upsilon_m=18.42\times10^{-6}m^2/s, \lambda_m=0.02814W/(m\cdot K), \mathrm{Pr}_m=0.71$$

故有：$(Gr\,\mathrm{Pr})_m=\frac{g\beta\Delta th^3}{\upsilon^2}\mathrm{Pr}_m \qquad \beta=\frac{1}{T_m}$

$$=\frac{9.81\times(90-20)\times(0.5)^3\times0.71}{(273+55)\times(18.42\times10^{-6})^2}=5.48\times10^8$$

查表得：$c=0.59, n=1/4$

$$\alpha=0.59\frac{\lambda_m}{h}(Gr\,\mathrm{Pr})_m^{1/4}=0.59\times\frac{0.02814}{0.5}\times(5.48\times10^8)^{1/4}=5.08W/(m^2\cdot K)$$

42.已知：$d=0.3m, t_w=450^\circ C, t_f=30^\circ C$

求：q_l

解：按$t_m=\frac{1}{2}(t_f+t_w)=240^\circ C$查表得：

$$\upsilon_m=39.86\times10^{-6}m^2/s, \lambda_m=0.0414W/(m\cdot K), \mathrm{Pr}_m=0.681$$

故有：$(Gr\,\mathrm{Pr})_m=\frac{g\beta\Delta td^3}{\upsilon_m^{\ 2}}\mathrm{Pr}_m=9.3\times10^7$

查表得：$c=0.125, n=1/3$

$\alpha=0.125\frac{\lambda_m}{d}(Gr\ Pr)_m^{1/3}$

$q_l=\alpha\pi d(t_w-t_f)=0.125(Gr\ Pr)_m^{1/3}\pi(t_w-t_f)=30W$

43.同 **42** 题的解法。

解：$t_m=\frac{1}{2}(t_f+t_w)=59^\circ C$

$\lambda_m = 2.89\times 3\ \ 1W\quad m\cdot(K\upsilon_m),\qquad 1\ 8.\ 8\ 7m^2\ 1\ 0s\quad_m = /\ \ ,\ P$

$$\beta = \frac{1}{T_m} = 3.01\times\ 1^3 0 1/K$$

故有：$(Gr\,\mathrm{Pr})_m = \dfrac{g\beta_m \Delta t d^3}{\upsilon_m^{\ 2}}\mathrm{Pr}_m = 2\times 10^7$

查表得：$c = 0.125, n = 1/3$

$$\therefore Nu_m = 0.\ 1\ 2\ 6\ G\ r\ (\quad P^{1/3}_m$$

$$q_l = \alpha\pi d(t_w - t_f) = 0.\ 1\ 2\ 7\ 5\ \ (G\ r^{1/3}\ P_m\ r)_m\quad t_w\ -(\quad t_f = \ )\quad 2\ 5W$$

44.已知：$t_w = 13^\circ C, t_f = 25^\circ C, F = 4\times 5m^2$

求：α, Q

解：按 $t_m = \dfrac{1}{2}(t_f + t_w) = 19^\circ C$ 查表得：

$$\upsilon_m = 15.02\times 10^{-6} m^2/s, \lambda_m = 0.02563 W/(m\cdot K), \mathrm{Pr}_m = 0.7111$$

$$\beta_m = \frac{1}{T_m} = 3.\ 4\ 2\ 5\times\ 1\ 0^{-3}\ 1/K$$

故有：$(Gr\,\mathrm{Pr})_m = \dfrac{g\beta_m \Delta t l^3}{\upsilon_m^{\ 2}}\mathrm{Pr}_m \qquad l = \dfrac{1}{2}(\ 4+\ \ 5) = \ \ 4m$

即：$(Gr\,\mathrm{Pr})_m = \dfrac{9.81\times 3.425\times 10^{-3}\times(25-13)\times 4.5^3\times 0.711}{(15.02\times 10^{-6})^2} = 1.64\times 10^{11}$

查表得：$c = 0.\ 1\ 5, n = 1/3$

$$\alpha = 0.15\frac{\lambda_m}{l}(Gr\,\mathrm{Pr})_m^{1/3} = 0.15\times\frac{0.02563}{4.5}\times(1.64\times 10^{11})^{1/3} = 4.7 W/(m^2\cdot K)$$

$$Q = \alpha F(t_f - t_w) = 4.7\times 4\times 5\times(25-13) = 1128W$$

45.略

46.已知：$d = 5.5m, t_w = 355^\circ C, t_f = 35^\circ C$

$(1\ t)_{w1} = \ \ 8\ 0C\quad t_{f1} = \ \ ^\circ 2\ 0C\quad ;(t_2 =)\ \ ^\circ\ \ \ 8\ 0t_2 = ,^\circ$

求：d_1, d_2

解： 按 $t_m=\frac{1}{2}(t_f+t_w)=195^{\circ}C$ 查表得：

$$\upsilon_m=34.26\times10^{-6}m^2/s,\lambda_m=3.893\times10^{-2}W/(m\cdot K),\mathrm{Pr}_m=0.68$$

有：$(Gr\,\mathrm{Pr})_m=\frac{g\beta_m\Delta tl^3}{\upsilon_m^{\ 2}}\mathrm{Pr}_m=\frac{9.81\times(355-35)\times5.5^3}{(273+195)\times(34.26\times10^{-6})^2}\times0.68=6.46\times10^{11}$

按 $t_{m1}=\frac{1}{2}(t_{f1}+t_{w1})=50^{\circ}C$ 查表（空气）得：

$$\upsilon_{m1}=17.95\times10^{-6}m^2/s,\lambda_m=2.83\times10^{-2}W/(m\cdot K),\mathrm{Pr}_{m1}=0.698$$

故有：$(Gr\,\mathrm{Pr})_{m1}=\frac{g\beta_{m1}\Delta tl^3}{\upsilon_{m1}^{\ 2}}\mathrm{Pr}_{m1}$

根据自模化有：$(Gr\,\mathrm{Pr})_m=(Gr\,\mathrm{Pr})_{m1}$

得：$d_1=\left[\frac{\upsilon_{m1}^{\ 2}}{g\beta_{m1}\Delta t_1\,\mathrm{Pr}_{m1}}(Gr\,\mathrm{Pr})_m\right]^{1/3}$

$$=\left[\frac{(273+50)\times(17.95\times10^{-6})^2}{9.81\times(80-20)\times0.698}\times6.46\times10^{11}\right]^{1/3}=5.47m$$

再按 $t_{m2}=\frac{1}{2}(t_{f2}+t_{w2})=50^{\circ}C$ 查表（水）得：

$$\upsilon_{m2}=0.556\times10^{-6}m^2/s,\mathrm{Pr}_{m2}=3.54,\beta_{m2}=4.57\times10^{-4}\ 1/K$$

同理得：$d_2=\left[\frac{\upsilon_{m2}^{\ 2}}{g\beta_{m2}\Delta t_2\,\mathrm{Pr}_{m2}}(Gr\,\mathrm{Pr})_m\right]^{1/3}$

$$=\left[\frac{(0.556\times10^{-6})^2}{9.81\times(80-20)\times3.54\times(4.57\times10^{-4})}\times6.46\times10^{11}\right]^{1/3}=0.594m$$

由此可以看出：方案 2 可以大大减小模型的尺寸。

47.已知：$h=650mm,t_f=15^{\circ}C,\alpha=4.82W/(m^2\cdot K),F=2m^2$

求：t_w,Q

解：设 $t_w=65^{\circ}C$，按 $t_m=\frac{1}{2}(t_f+t_w)=40^{\circ}C$ 查表得：

$\lambda_m=2.76\times10^{-2}W/(m\cdot K)$，$\upsilon_m=16.96\times10^{-6}m^2/s$，Pr

$(Gr\,\mathrm{Pr})_m=\frac{g\beta_m\Delta th^3}{\upsilon_m^{\ 2}}$ $\mathrm{Pr}_m=\frac{9.81\times(65-15)\times0.65^3}{(16.96\times10^{-6})^2\times313}\times0.699\times1.0^{9}$

查表有：$c=0.15, n=1/3$

$$\alpha' = 0.1\frac{\lambda_m}{h}(Gr\,\mathrm{Pr})_m^{1/3} = 0.1\times\frac{2.76\times10^{-2}}{0.65}\times(1.04\times10^{9})^{1/3} = 4.28W/(m^2\cdot K)\neq\alpha$$

重新设$t_w = 85^\circ C$,按$t_m = \frac{1}{2}(t_f + t_w) = 50^\circ C$查表得：

$\lambda_m = 2.83\times10^{-2}W/(m\cdot K)$, $\upsilon_m = 17.95\times10^{-6}m^2/s$, $\mathrm{Pr}_m = 1.02$

$$(Gr\,\mathrm{Pr})_m = \frac{g\beta_m\Delta t h^3}{\upsilon_m^2}\mathrm{Pr}_m = 1.26\times10^{9}$$

$$\alpha' = 0.1\frac{\lambda_m}{h}(Gr\,\mathrm{Pr})_m^{1/3} = 4.72W/(m^2\cdot K)\approx\alpha$$

$$\therefore t_w = 85^\circ C$$

再：$Q = \alpha F(t_W - t_F) = 4.82\times2\times(85-15) = 674W$

49-55.略

56.已知： $h = 2.5m, t_w = 30^\circ C, t_F = 10^\circ C$

求：$u_{\max}, u_{\min}$

解：当$\frac{Gr}{\mathrm{Re}^2}\geq0.1$时，不可以忽略自然对流的影响

即：$$\frac{\frac{g\beta_m\Delta t h^3}{\upsilon^2}}{\left(\frac{u_\infty h}{\upsilon}\right)^2} = \frac{g\beta_m\Delta t h}{u_\infty^2}\geq0.1, \beta_m = \frac{1}{T_m}, t_m = \frac{1}{2}(t_w + t_f)$$

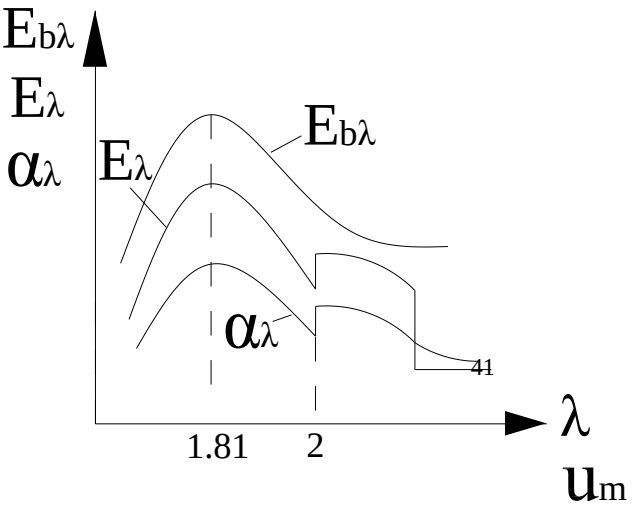
$$\therefore u_{max} = u_\infty \leq \left(\frac{g\beta_m\Delta t h}{0.1}\right)^{1/2} = \left(\frac{9.81\times(30-10)\times2.5}{0.1\times(273+20)}\right)^{1/2} = 4.1m/s$$

而当$\frac{Gr}{\mathrm{Re}^2}\geq10$时，可以忽略受迫对流的影响

即：

$$u_{min} = u_\infty \leq \left(\frac{g\beta_m\Delta t h}{10}\right) = 0.41m/s$$

57.略



第八章　热辐射的基本定律

1.略　2.略　3.略（可见光与红外线的波长不同）　4.略

5.对于一般物体，在热平衡条件下，根据基尔

霍夫定律有：$\alpha_\lambda = \varepsilon_\lambda = \dfrac{E_\lambda(\lambda,T)}{E_{b\lambda}(\lambda,T)}$，即：$E_\lambda(\lambda,T) = \alpha_\lambda E_{b\lambda}(\lambda,T)$（如右图）

6. 略　7. 略

8. 已知：$\lambda = 0.4 \sim 2.5\mu m, T_1 = 1500K$

$T_2 = 2000K, T_3 = 6000K$

求：$F_{b(\lambda_1\sim\lambda_2)_1}, F_{b(\lambda\sim\lambda_2)_2}, F_{b(\lambda_1\sim\lambda_2)_3}$

解：$\begin{cases}\lambda_1 T_1 = 0.4\times 1500 = 600\mu m\cdot K\\ \lambda_2 T_1 = 2.5\times 1500 = 3750\mu m\cdot K\end{cases}$

查表得：$F_{b(0\sim\lambda_1)_1} = 0, F_{b(0\sim\lambda_2)_1} = 0.4335$

即：$F_{b(\lambda_1\sim\lambda_2)_1} = F_{b(0\sim\lambda_2)_1} - F_{b(0\sim\lambda_1)_1} = 0.4335$

同理：$\begin{cases}\lambda_1 T_2 = 0.4\times 2000 = 800\mu m\cdot K\\ \lambda_2 T_2 = 2.5\times 2000 = 5000\mu m\cdot K\end{cases}$

查表得：$F_{b(0\sim\lambda_1)_2} = 0, F_{b(0\sim\lambda_2)_2} = 0.6338$

即：$F_{b(\lambda_1\sim\lambda_2)_2} = F_{b(0\sim\lambda_2)_2} - F_{b(0\sim\lambda_1)_2} = 0.6338$

再：$\begin{cases}\lambda_1 T_3 = 0.4\times 6000 = 2400\mu m\cdot K\\ \lambda_2 T_3 = 2.5\times 6000 = 15000\mu m\cdot K\end{cases}$

查表得：$F_{b(0\sim\lambda_1)_3} = 0.1403, F_{b(0\sim\lambda_2)_3} = 0.9690$

即：$F_{b(\lambda_1\sim\lambda_2)_3} = F_{b(0\sim\lambda_2)_3} - F_{b(0\sim\lambda_1)_3} = 0.8287$

9. 已知：$\lambda_{\max} = 2\mu m, \lambda_{1\sim 2} = 1 \sim 4\mu m$

求：$F_{b1\sim}$

解：有维恩位移定律 $\lambda_{max}T = 2897$，有：$T = \dfrac{2897.6}{\lambda_{max}} = \dfrac{2897.6}{2} = 1448K$

再 $\begin{cases}\lambda_1 T = 1\times 1448 = 1448\mu m\cdot K\\ \lambda_2 T = 4\times 1448 = 5792\mu m\cdot K\end{cases}$

查表有：$F_{b(0\sim\lambda_1)} = 0.0103, F_{b(0\sim\lambda_2)} = 0.72$

即：$F_{b(\lambda_1\sim\lambda_2)} = F_{b(0\sim\lambda_2)} - F_{b(0\sim\lambda_1)} = 0.72 - 0.0103 = 0.7097$

10. 已知：$T_1 = 2000K, \lambda = 0.38 \sim 0.76\mu m, T_2 = 5762K$

求：$F_{b(\lambda_1 \sim \lambda_2)_1}, F_{b(\lambda_1 \sim \lambda_2)_2}$

解：$\begin{cases} \lambda_1 T_1 = 0.38 \times 2000 = 760\mu m \cdot K \\ \lambda_2 T_1 = 0.76 \times 2000 = 1520\mu m \cdot K \end{cases}$

查表得：$F_{b(0 \sim \lambda_1)_1} = 0, F_{b(0 \sim \lambda_2)_1} = 0.0149$

即：$F_{b(\lambda_1 \sim \lambda_2)_1} = F_{b(0 \sim \lambda_2)_1} - F_{b(0 \sim \lambda_1)_1} = 0.0149$

再：$\begin{cases} \lambda_1 T_2 = 0.38 \times 5762 = 2190\mu m \cdot K \\ \lambda_2 T_2 = 0.76 \times 5672 = 4379\mu m \cdot K \end{cases}$

查表得：$F_{b(0 \sim \lambda_1)_2} = 0.0991, F_{b(0 \sim \lambda_2)_2} = 0.5454$

即：$F_{b(\lambda_1 \sim \lambda_2)_2} = F_{b(0 \sim \lambda_2)_2} - F_{b(0 \sim \lambda_1)_2} = 0.446$

11. 已知：$F_1 = 1cm^2, I_n = 3500W/(m^2 \cdot sr), F_2 = F_3 = F_4 = F_1, r = 0.5m, \varphi_2 = -60^\circ$，

$\varphi_3 = 0^\circ, \varphi_4 = 45^\circ$

求：$(1) I_2, I_3, I_4$；$(2) \omega_2, \omega_3, \omega_4$；$(3) Q_2, Q_3, Q_4$

解：（1） $I_2 = I_3 = I_4 = I_n = 3500W/(m^2 \cdot sr)$

（2） $\omega_2 = \dfrac{F_2}{r^2} = \dfrac{1 \times 10^{-4}}{0.5^2} = 4 \times 10^{-4} sr$

由 $F_2 = F_3 = F_4 = F$ 知：$\omega_2 = \omega_3 = \omega_4 = 4 \times 10^{-4} sr$

（3） 由 $I_n = \dfrac{Q_n}{F_1 \cos\varphi\omega}$ 知：$Q_n = I_n F_1 \cos\varphi\omega$

$\therefore Q_2 = I_2 F_2 \cos\varphi_2 \omega_2 = 3500 \times 1^{-4}10 \cos(-60^\circ) 4 = 4 \times 10^{-5} W$

$Q_3 = I_3 F_3 \cos\varphi_3 \omega_3 = 1 \times 4^{-4}10 W$

$Q_4 = I_4 F_4 \cos\varphi_4 \omega_4 = 9.9 \times 10^{-5} W$

12.已知：$\lambda_{1\sim2} = 0.4 \sim 2.5\mu m, \tau = 0.95, T_1 = 5762K, T_2 = 273 + 50K$

求：$F_{(\lambda_1 \sim \lambda_2)_1}, F_{(\lambda_1 \sim \lambda_2)_2}$

解：$\begin{cases}\lambda_1 T_1 = 0.4\times 5762 = 2304.8\mu m\cdot K\\ \lambda_2 T_1 = 2.5\times 5762 = 14405\mu m\cdot K\end{cases}$

查表得：$F_{b(0\sim\lambda_1)_1} = 0.1206, F_{b(0\sim\lambda_2)_1} = 0.9654$

$\therefore F_{(\lambda_1\sim\lambda_2)_1} = \tau F_{b(\lambda_1\sim\lambda_2)_1} = \tau(F_{b(0\sim\lambda_2)_1} - F_{b(0\sim\lambda_1)_1}) = 0.95\times(0.9654-0.1206) = 0.803$

再：$\begin{cases}\lambda_1 T_2 = 0.4\times 323 = 129.2\mu m\cdot K\\ \lambda_2 T_2 = 2.5\times 323 = 807.5\mu m\cdot K\end{cases}$

查表得：$F_{b(0\sim\lambda_1)_2} = 0, F_{b(0\sim\lambda_2)_2} = 0$

$\therefore F_{(\lambda_1\sim\lambda_2)_2} = \tau F_{b(\lambda_1\sim\lambda_2)_2} = 0$

13. 已知：$T = 1500K, \varepsilon_{\lambda(0\sim1)} = 0.1, \varepsilon_{\lambda(1\sim3)} = 0.4, \varepsilon_{\lambda(3\sim5)} = 0.2$

求：ε, E

解：$E = \int_0^\infty \varepsilon_\lambda E_{b\lambda} d\lambda = \int_0^1 \varepsilon_{(\lambda 0\sim 1)} E_{b\lambda} d\lambda + \int_1^3 \varepsilon_{\lambda(1\sim 3)} E_{b\lambda} d\lambda + \int_3^5 \varepsilon_{\lambda(3\sim 5)} E_{b\lambda}$

$= \left[\varepsilon_{\lambda(0\sim1)} F_{b(0\sim1)} + \varepsilon_{\lambda(1\sim3)} F_{b(1\sim3)} + \varepsilon_{\lambda(3\sim5)} F_{b(3\sim5)}\right] E_b$

而：$\begin{cases}\lambda_0 T = 0\times 1500 = 0\\ \lambda_2 T = 3\times 1500 = 4500\\ \lambda_3 T = 5\times 1500 = 7500\\ \lambda_1 T = 1\times 1500 = 1500\end{cases}\mu m\cdot K$

查表有：$F_{b(0\sim\lambda_0)} = 0, F_{b(0\sim\lambda_1)} = 0.0138, F_{b(0\sim\lambda_2)} = 0.5641, F_{b(0\sim\lambda_3)} = 0.8344$

即：$F_{b(0\sim 1)} = F_{b(0\sim\lambda_1)} - F_{b(0\sim\lambda_0)} = 0.0138$

$F_{b(1\sim3)} = F_{b(0\sim\lambda_2)} - F_{b(0\sim\lambda_1)} = 0.5641 - 0.0138 = 0.5503$

$F_{b(3\sim 5)} = F_{b(0\sim\lambda_3)} - F_{b(0\sim\lambda_2)} = 0.8344 - 0.5641 = 0.2703$

$\therefore E = (0.1\times 0.0138 + 0.4\times 0.5503 + 0.2\times 0.2703)\times 5.67\times\left(\frac{1773}{100}\right)^4$

$= 1558 W/m^2$

$\varepsilon = \frac{E}{E_b} = (0.1\times 0.0138 + 0.4\times 0.5503 + 0.2\times 0.2703$

14.已知：$\alpha_{\lambda(0\sim6)}=0.2,\alpha_{\lambda(6\sim10)}=\frac{0.7}{4}\lambda-0.85,\alpha_{\lambda(10\sim\infty)}=0.9,G_{\lambda(0\sim6)}=\frac{400}{6}\lambda W/(m^2\cdot\mu m)$

$G_{\lambda(6\sim10)}=400W/(m^2\cdot\mu m)$ ， $G_{\lambda(10\sim12)}=2400-200\lambda W/(m^2\cdot\mu m)$

求：α

解：$\alpha=\dfrac{\int_0^\infty\alpha_\lambda G_\lambda d_\lambda}{\int_0^\infty G_\lambda d_\lambda}$

$$=\frac{\int_0^6\alpha_{\lambda(0\sim6)}G_{\lambda(0\sim6)}d+\int_6^{10}\alpha_{\lambda(6\sim10)}G_{\lambda(6\sim10)}d+\int_{10}^{12}\alpha_{\lambda(10\sim12)}G_{\lambda(10\sim12)}d}{\int_0^6 G_{\lambda(0\sim6)}d+\int_6^{10}G_{\lambda(6\sim10)}d+\int_{10}^{12}G_{\lambda(10\sim12)}d}=0.4$$

15.已知：$T_1=500K,\lambda\le2\mu m,\alpha_{\lambda(0\sim2)}=0.9,\lambda>2\mu m,\alpha_{\lambda(2\sim\infty)}=0.1,$

$T_2=800K,T_3=5800K$

求：$\varepsilon_1,\alpha_2,\alpha$

解：$\varepsilon_1=\dfrac{E_1}{E_{b1}}=\dfrac{\int_0^2\alpha_{\lambda(0\sim2)}E_{b_1\lambda}d_\lambda+\int_2^\infty\alpha_{\lambda(2\sim\infty)}E_{b_1\lambda}d_\lambda}{E_{b1}}$

$=\alpha_{\lambda(0\sim2)}F_{b_1(0\sim2)}+\alpha_{\lambda(2\sim\infty)}F_{b_1(2\sim\infty)}$

$=\alpha_{\lambda(0\sim2)}F_{b_1(0\sim2)}+\alpha_{\lambda(2\sim\infty)}F_{b_1(2\sim\infty)}-\alpha_{\lambda(0\sim2)}F_{b_1(}$

$=(\alpha_{\lambda(0\sim2)}-\alpha_{\lambda(2\sim\infty)})F_{b_1(0\sim2)}+\alpha_{\lambda(2\sim\infty)}F_{b_1(0\sim\infty)}$

而：$\lambda T_1=2\times500=1000\mu m\cdot K$,查表得：$F_{b(0\sim2)}=3\times10^{-4}$

$\therefore\varepsilon_1=(0.9-0.1)\times3\times10^{-4}+0.1=$

$\alpha_2=\dfrac{\int_0^2\alpha_{\lambda(0\sim2)}E_{b_2\lambda}d_\lambda+\int_2^\infty\alpha_{\lambda(2\sim\infty)}E_{b_2\lambda}d_\lambda}{E_{b_2}}$

$=(\alpha_{\lambda(0\sim2)}-\alpha_{\lambda(2\sim\infty)})F_{b_2(0\sim2)}+\alpha_{\lambda(2\sim\infty)}F_{b_2(0\sim\infty)}$

而：$\lambda T_2=2\times800=1600\mu m\cdot K$,查表得：$F_{b_2(0\sim2)}=0.0197$

$\therefore\alpha_2=(0.9-0.1)\times0.0197+0.1\times1=$

同理：$\lambda T_3=2\times5800=11600\mu m\cdot K$,查表得：$F_{b_3(0\sim2)}=0.94$

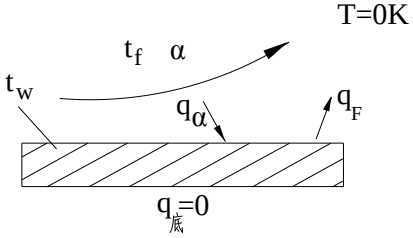
$\therefore\alpha_3=(0.9-0.1)\times0.94+0.1\times1=0.852$

16.同上题，略

17.略

18.已知：$T=0K, t_f=3℃, \alpha=15W/(m^2\cdot K), \varepsilon_1=0.9, \varepsilon_2=0.1$

求：t_{w1}, t_{w2}



解：热平衡时有：$q_\alpha = q_F$

即：$\alpha(t_f - t_w) = \varepsilon C_b \left(\frac{T_w}{100}\right)^4$

当$\varepsilon_1=0.9$时，有：$15\times(3-t_{w1}) = 0.9\times 5.67\times\left(\frac{273+t_{w1}}{100}\right)^4$

试凑可得：$t_{w1} = -10.3℃$

当$\varepsilon_2=0.1$时，有：$15\times(3-t_{w2}) = 0.1\times 5.67\times\left(\frac{273+t_{w2}}{100}\right)^4$

试凑可得：$t_{w2} = 0.9℃$

第九章：辐射换热计算

1.略

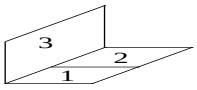
2.略

3.略

4.略

5.已知：t_1=1000℃、t_2=500℃、t_2'=700℃、$\varepsilon_1=0.8$、$\varepsilon_2=0.5$、$Z_{1,2}=1$

求：$q_{1,2}$、$q'_{1,2}$、$q''_{1,2}$



解：$q_{1,2} = E_{b_1} - E_{b_2}$

$$= C_b\left[(\frac{T_1}{100})^4 - (\frac{T_2}{100})^4\right]$$

$$= 5.67\times\left[(\frac{273+1000}{100})^4 - (\frac{273+500}{1000})^4\right] = 128.7\,kw/m^2$$

同理：$q'_{1,2} = 5.67\times\left[(\frac{273+1000}{1000})^4 - (\frac{273+700}{100})^4\right] = 98.1\,kw/m^2$

再：$q_{1,2}'' = -\dfrac{E_{b_1}-E_{b_2}}{\dfrac{1}{\varepsilon_1}+\dfrac{1}{\varepsilon_2}-1} = \dfrac{q_{1,2}}{\dfrac{1}{\varepsilon_1}+\dfrac{1}{\varepsilon_2}-1} = \dfrac{128.7}{\dfrac{1}{0.8}+\dfrac{1}{0.5}-1} = 57.2\ kw/m^2$

6.略

7.略

8.（1） $F_3 Z_{3,(1+2)} = F_3 Z_{3,1} + F_3 Z_{3,2}$

$\therefore Z_{3,(1+2)} = Z_{3,1} + Z_{3,2}$

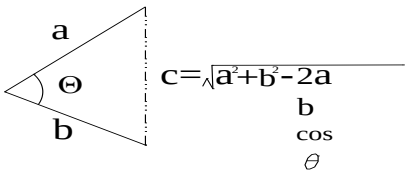
(2) $F_{(1+2)}, Z_{(1+2),3} = F_3 Z_{3,(1+2)} = F_3 Z_{3,1} + F_3 Z_{3,2}$

而：$F_3 Z_{3,1} = F_1 Z_{1,3}$，$F_3 Z_{3,2} = F_2 Z_{2,3}$

$\therefore F_{(1+2)}$，$Z_{(1+2),3} = F_1 Z_{1,3} + F_2 Z_{2,3}$

即：$Z_{(1+2),3} = \dfrac{1}{F_{(1+2)}}(F_1 Z_{1,3} + F_2 Z_{2,3})$

$\therefore Z_{(1+2),3} \neq Z_{1,3} + Z_{2,3}$

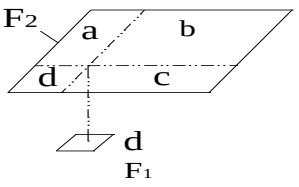


9．略（ $Z_{a,b} = \dfrac{a+b-\sqrt{a^2+b^2-2ab\cos\theta}}{2a}$ **）**

10.（1）、 $Z_{dF_1,F_a} + Z_{dF_1,F_b} + Z_{dF_1,F_c} + Z_{dF_1,F_d}$

而：Z_{dF_1,F_a}、Z_{dF_1,F_b}、Z_{dF_1,F_c}、Z_{dF_1,F_d} 可查图（P_{242}　图 **9-19**）得

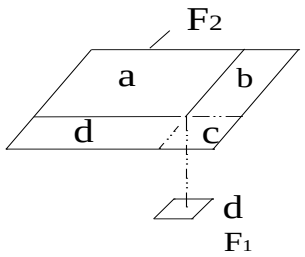
结果略



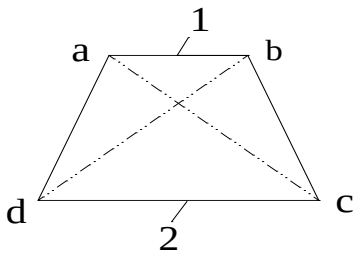
（2）$Z_{1,a}=Z_{1,(a+b+c+d)}-Z_{1,(b+c)}-Z_{1,(c+d)}+Z_{1,c}$

而$Z_{1,(a+b+c+d)}$、$Z_{1,(b+c)}$、$Z_{1,(c+d)}$、$Z_{1,c}$可查图（P_{242}　图 9-19）得

结果略



（3）$Z_{1,2}=Z_{ab,cd}=\dfrac{(l_{bd}+l_{ac})-(l_{ad}+l_{bc})}{2l_{ab}}$　　结果略



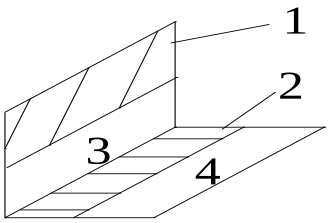
（4）$Z_{2,(1+3)}=Z_{2,3}+Z_{2,1}$

而：$F_1Z_{1,2}=F_2Z_{2,1}$

即：$Z_{2,1}=\dfrac{F_2}{F_1}Z_{1,2}$　　由此可得：

$$Z_{1,2}=\frac{F_2}{F_1}(Z_{2,(1+3)}-Z_{2,3})$$

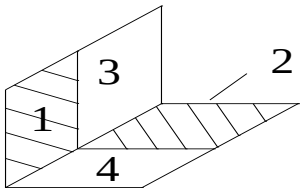
而：$Z_{2,(1+3)}$，$Z_{2,3}$　可查图（P_{243}　图 9-20）得　结果略



（5）$F_{(1+3)}Z_{(1+3),(2+4)} = F_1Z_{1,4} + F_1Z_{1,2} + F_3Z_{3,2} + F_3Z_{3,4}$

而：$F_1 = F_2 = F_3 = F_4$，既有：$Z_{1,4} = Z_{3,2}$，$Z_{1,2} = Z_{3,4}$

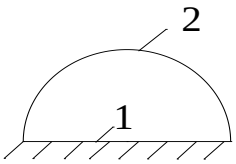
整理有：$Z_{1,2} = Z_{(1+3)，(2+4)} - Z_{1,4}$　　结果略



（6）$Z_{2,1} + Z_{2,2} = 1$

而　　$Z_{2,1} = \frac{F_1}{F_1}Z_{1,2}$，$Z_{1,2} = 1$

$\therefore Z_{2,2} = 1 - F_1 / F_2$



11.(1) $Z_{1,2} = 0$

(2) $Z_{1,3} = \frac{F_3}{F_1}(Z_{3,(1+2)}, -Z_{3,2})$

(3) $Z_{1,4} = Z_{1,4a} + Z_{1,4b}$

而：$Z_{1,4b}\cdot F_1=\frac{1}{2}(F_{(1+2)}Z_{(1+2),4}-F_1Z_{1,4a}-F_2Z_{2,4b})$

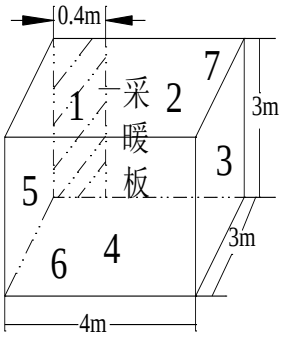
即：$Z_{1,4}=\frac{1}{2F_1}(F_{(1+2)}Z_{(1+2),4}-F_1Z_{1,4a}-F_2c)$

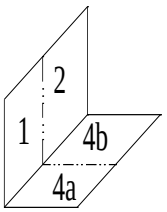
$Z_{(1+2),4}$、$Z_{1,4a}$、$Z_{1,4a}$ 可查图得（略）

（4）$Z_{1,5}$ 查图即可得（略）

（5）$Z_{1,7}=Z_{1,4}$

（6）$Z_{1,6}=1-\sum_{\substack{i=1\\(i\neq 6)}}^{7}Z_{1,i}$ （略）





12.略

13.已知： $d=100mm$，$t_{w_1}=100℃$，$t_{w_2}=27℃$，$\varepsilon_1=0.85$

求： q_l

解： $\because Z_{1,2}=1,\ F_1/F_2\leq 1$, 有：$\varepsilon_s=\varepsilon_1$

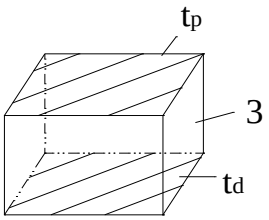
$\therefore q_l\frac{Q}{L}=\varepsilon_1\pi d(E_{b_1}-E_{b_2})$

$$=5.67\times 0.1\pi\left[(\frac{100+273}{100})^4-(\frac{27+273}{100})^4\right]=201\,w/m$$

14.略

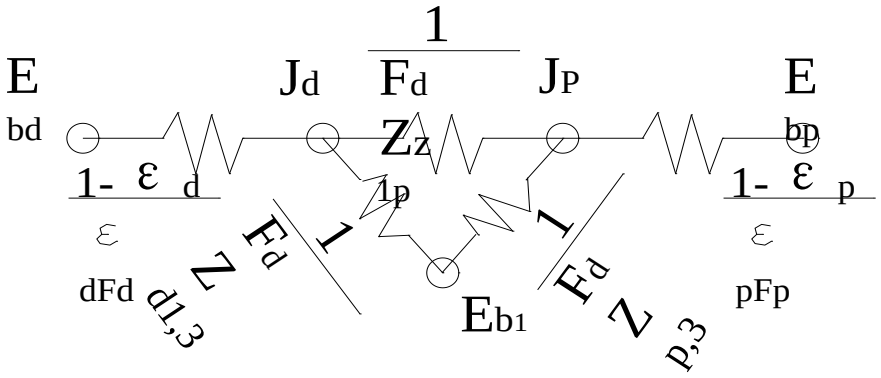
15.已知： $a\times b\times c=3\times 4\times 2.5=30m^3, t_d=27℃, t_p=12℃, q_3=0, \varepsilon=0.8$

求： $Q_{d,p}, t_3$



解： $\because \varepsilon_d = \varepsilon_p = \varepsilon_3 = \varepsilon, F_p = F_d$

$\therefore \dfrac{1-\varepsilon_d}{\varepsilon_d F_d} = \dfrac{1-\varepsilon_p}{\varepsilon_p F_p}, Z_{d,p}$ **查图即得（** P_{242} **图 9-20）**



$Z_{d,3} = 1 - Z_{d,d} - Z_{d,p}$ **而** $Z_{dd} = 0$

即 $Z_{d,3} = 1 - Z_{d,p}$

$Z_{p,3} = Z_{d,3} (\because F_p = F_d)$

$$\therefore \varepsilon_3 = \frac{1}{2 \times \dfrac{1-\varepsilon}{\varepsilon F_d} + \left(\dfrac{1}{\dfrac{1}{\dfrac{1}{F_d Z_{d,3}} + \dfrac{1}{F_p Z_{p,3}}} + \dfrac{1}{F_d Z_{d,p}}} \right)} = \frac{1}{2 \times \dfrac{1-\varepsilon}{\varepsilon F_d} + 2 \times \dfrac{1}{F_d (1 + Z_{d,p})}}$$

$Q_{d,p} = \varepsilon_s (E_{bd} - E_{bp})$ **(结果略)**

再: $J_d = E_{bd} - \frac{1-\varepsilon_d}{\varepsilon_d F_d} Q_{d,p}, J_P = E_{bp} - \frac{1-\varepsilon_p}{\varepsilon_p F_p} Q_{p,d}$

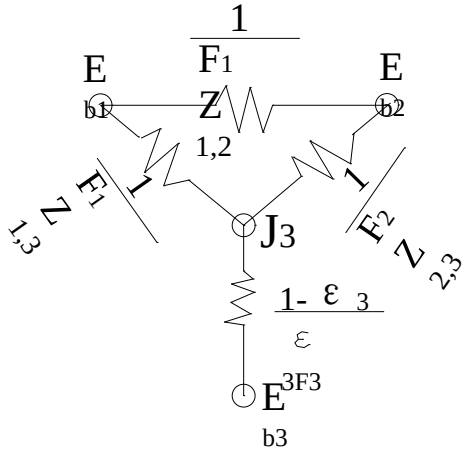
对节点 d 有： $\frac{E_{bd} - J_d}{\frac{1-\varepsilon_d}{\varepsilon_d F_d}} + \frac{J_p - J_d}{\frac{1}{Z_{d,p} F_d}} + \frac{J_3 - J_1}{\frac{1}{Z_{d,3} F_d}} = 0$

既有: $J_3 = J_1 - \frac{1}{F_d Z_{d,3}} \left(\frac{E_{db} - J_d}{\frac{1-\varepsilon_d}{\varepsilon_d F_d}} + \frac{J_p - J_d}{\frac{1}{F_d Z_{d,p}}} \right)$

而： $E_{b3} = J_3$　　即： $c_b T_3^4 = J_3, T_3 = \left(\frac{J_3}{c_b}\right)^{1/4}$　　（结果略）

16.已知： $F = 1 \times 1 = 1m^2, D = 2m, t_1 = 400℃, t_2 = 123℃ t_w = 20℃$， $\varepsilon_1 = \varepsilon_2 = 1$

求： $Q_{1,2}, Q_{1,3}, Q_{2,3}$



解：由： $\frac{Z}{D} = Y_2, \frac{Y}{D} = \frac{1}{2}$　　查图知

$Z_{1,2} = Z_{2,1} = 0.07$

再由完整性知： $Z_{1,3} = 1 - Z_{1,2} = 0.93$

$Z_{2,3} = 1 - Z_{2,1} = 0.93$

$\therefore Q_{1,2} = \frac{E_{b_1} - E_{b_3}}{\frac{1}{F_1 Z_{1,3}}} = F_1 Z_{1,2} (E_{b_1} - E_{b_2})$　　（结果略）

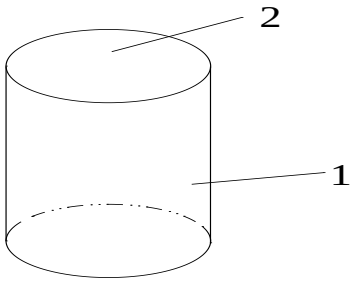
又因：$F_3 >> F_1, F_3 >> F_2$，可取其表面热阻$\frac{1-\varepsilon_3}{\varepsilon_3 F_3} \approx 0$,即：$E_{b_3} = J_3$

故有：$Q_{1,3} = \frac{E_{b_1} - E_{b_3}}{\frac{1}{F_1 Z_{1,3}}} F_1 Z_{1,3}(E_{b_1} - E_{b_3})$

$$Q_{2,3} = \frac{E_{b_2} - E_{b_3}}{\frac{1}{F_2 Z_{2,3}}} F_2 Z_{2,3}(E_{b_2} - E_{b_3})$$

17.已知：$d = h = 100mm, t_1 = 327℃, \varepsilon_1 = 0.75, t_2 = 27℃$

求：$Q_{1,2}$

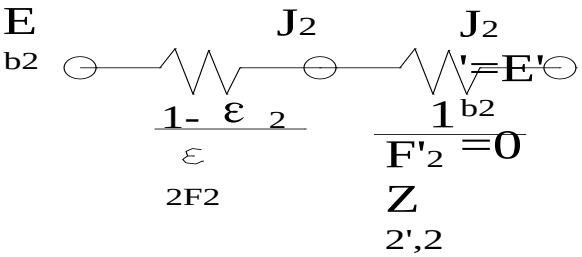


解：设上表面为$2'$，其$\varepsilon_{2'} = 1$，$T_{2'} = 0K$,其与 1 面间的辐射换热量为：

$$Q_{1,2'} = \frac{E_{b_1} - E_{b_{2'}}}{\frac{1-\varepsilon_1}{\varepsilon_1 F_1} + \frac{1}{F_1 Z_{1,2'}}} = \frac{E_{b_1}}{\frac{1-\varepsilon_1}{\varepsilon_1 F_1} + \frac{1}{F_{2'}}}$$

$$F_1 = \pi dh + \frac{\pi}{4} d^2$$

$$F_{2'} = \frac{\pi}{4} d^2$$



再$2'$面与房间间的辐射换热为：

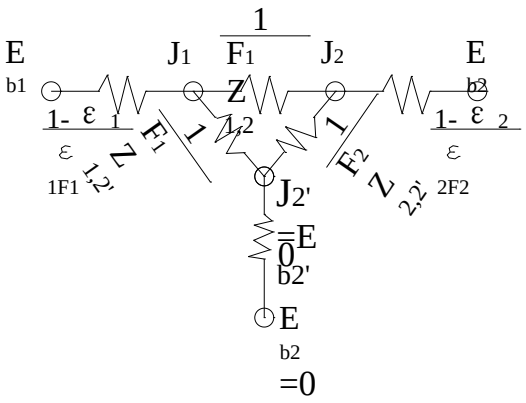
而：$Z_{2',2} = 1, F_2 >> F_{2'}$

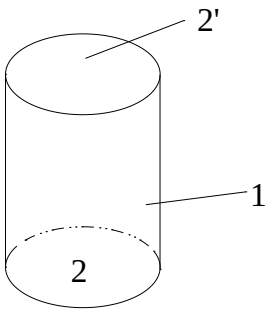
故： $Q_{2,2'}=\dfrac{E_{b_2}-E_{b_{2'}}}{\dfrac{1-\varepsilon_2}{\varepsilon_2 F_2}+\dfrac{1}{F_{2'}Z_{2',2}}}=F_{2'}E_{b_2}$

热损失 $Q_{1,2}=Q_{1,2'}-Q_{2,2'}$

$$=\frac{E_{b_1}}{\dfrac{1-\varepsilon_1}{\varepsilon_1 F_1}+\dfrac{1}{F_{2'}}}-F_{2'}E_{b_2}=50.5w$$

18.





计算略， $2'$ 面与房间间的辐射换人量同上题，结果略

19.略

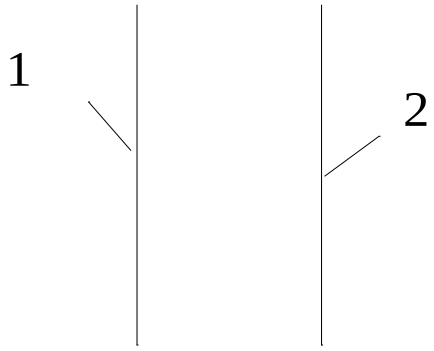
20.略

21.已知：$\varepsilon_1=\varepsilon_2=0.4$，$\varepsilon_3=0.04$，$t_1=250℃$，$t_2=40℃$

求：$q_{1,2},q'_{1,2},t_3$

无遮热板时：$F_1=F_2,Z_{1,2}=1$

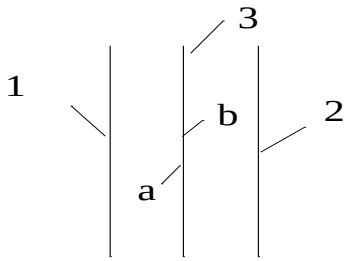
$q_{1,2}=\dfrac{E_{b_1}-E_{b_2}}{\dfrac{1}{\varepsilon_1}+\dfrac{1}{\varepsilon_2}-1}=$ （结果略）



有遮热板时：

$$q_{1,3a}=\frac{E_{b_1}-E_{b_{3a}}}{\frac{1}{\varepsilon_1}+\frac{1}{\varepsilon_2}-1} \qquad q_{3b,2}=\frac{E_{b_{3b}}-E_{b_2}}{\frac{1}{\varepsilon_3}+\frac{1}{\varepsilon_2}-1}$$

$T_{3a}=T_{3b},\varepsilon_1=\varepsilon_2$ 即： $E_{b_{3a}}=E_{b_{3b}}$



而 $q_{1,3a}=q_{3b,2}$ 即：

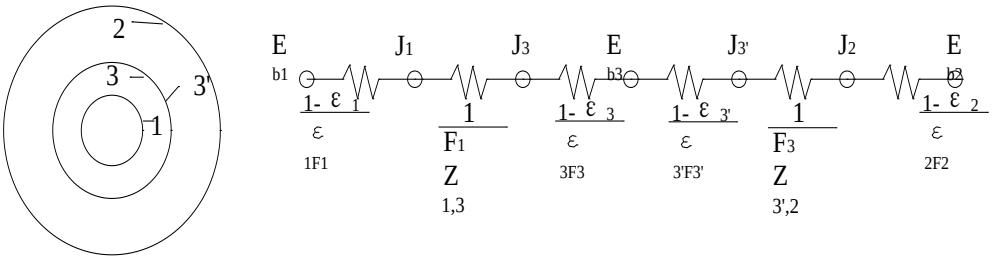
$$\frac{E_{b_1}-E_{b_{3a}}}{\frac{1}{\varepsilon_1}+\frac{1}{\varepsilon_2}-1}=\frac{E_{b_{3b}}-E_{b_2}}{\frac{1}{\varepsilon_3}+\frac{1}{\varepsilon_2}-1}\Rightarrow E_{b_3}=\frac{1}{2}(E_{b_1}+E_{b_2})$$

即：$T_3=\left[\frac{1}{2}\left(T_1^4+T_2^4\right)\right]^{1/4}=$ （结果略）

再 $q'_{1,2}=q_{1,3a}=q_{3b,2}$ 即：

$$q'_{1,2}=\frac{E_{b_1}-E_{b_3}}{\frac{1}{\varepsilon_1}+\frac{1}{\varepsilon_3}-1}=$$ （结果略）

22.



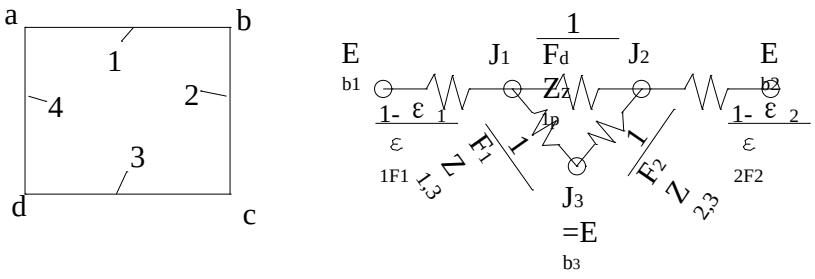
$\varepsilon_3 = \varepsilon_3^{'}, Z_{1,3} = 1, Z_{3^{'},2} = 1$　步骤同上题，计算结果略。

23.略

24.已知： $l_{ab} = l_{dc} = 4m, l_{bc} = l_{ad} = 3m, \varepsilon_2 = 0.6$

$$\varepsilon_1 = 0.8, t_1 = 1000K, t_2 = 500K, q_3 = 0$$

求： $q_{1,2}$



解： $Z_{1,2} = \dfrac{(l_{ac} + l_{bd}) - (l_{ad} + l_{bc})}{2l_{ab}}$

$Z_{2,1}=\frac{F_1}{F_2}Z_{1,2}, Z_{1,3}=1-Z_{1,2}, Z_{2,3}=1-Z_{2,1}$ **计算步骤同 15 题，计算略。**

25.已知： $d_1=200mm, d_2=300mm, t_1=280℃, t_f=30℃$

$$\alpha=25\,{}^{w}\!/_{(m^2\cdot K)}, \varepsilon_1=0.85, \varepsilon_2=0.3$$

求： $q_l, t_2, q_l^{'}$

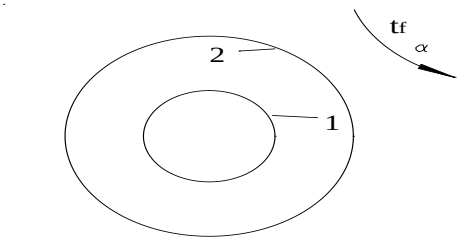
解：（1） $q_{lR}=\dfrac{E_{b_1}-E_{b_2}}{\dfrac{1-\varepsilon_1}{\varepsilon_1F_1}+\dfrac{1}{F_1Z_{1,2}}+\dfrac{1-\varepsilon_2}{\varepsilon_2F_2}}\cdot\dfrac{1}{l}$

$Z_{1,2}=1$

$F_1=\pi d_1 l$

$F_2=\pi d_2 l$

$$=\frac{(E_{b_1}-E_{b_2})}{\frac{1}{\varepsilon_1}+(\frac{1}{\varepsilon_2}-1)\frac{d_1}{d_2}}\cdot\pi d_1$$



而： $q_{lR}=q_{lc}=q_l$

即： $\alpha\pi d_2(t_2-t_{f_2})=\dfrac{\pi d_1(E_{b_1}-E_{b_2})}{\dfrac{1}{\varepsilon_1}+(\dfrac{1}{\varepsilon_2}-1)\dfrac{d_1}{d_2}}$

试凑得 t_2 **（结果略）**

（2） $q_l=\alpha\pi d_2(t_2-t_f)=$ **（结果略）**

（3） $q_l^{'}=q_{lR}^{'}+q_{lc}^{'}$

$=\pi d_1\alpha(t_1-t_f)+\pi d_1\varepsilon_1E_{b_1}$ **（** $q_{lR}^{'}$ **考虑气体及周围物体的辐射。）**

（结果略）

26.已知： $t_1=400℃, t_2=350℃, \varepsilon_1=0.8, \varepsilon_2=0.7, \alpha=35\,{}^{w}\!/_{(m^2\cdot K)}$

求： t_f, Δ

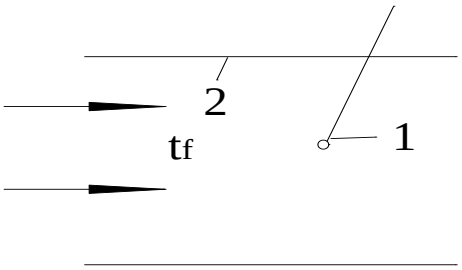
解：$F_2 / F_1 >> 1, Z_{1,2} = 1$　　1—热电偶

2—管壁

由热平衡知：

$$q_{1,2} = \varepsilon_1 (E_{b_1} - E_{b_2}) = q_c = \alpha (t_f - t_1)$$

即：$t_f = \frac{\varepsilon_1}{\alpha}(E_b - E_{b_2}) + t_1 =$　（结果略）



$$\Delta = \left| \frac{t_f - t_1}{t_f} \right| \times 100\% =$$（结果略）

讨论：$\alpha \uparrow \rightarrow (t_f - t_1) \downarrow \rightarrow \Delta \downarrow$

$\varepsilon_1 \uparrow \rightarrow (t_f - t_1) \uparrow \rightarrow \Delta \uparrow$

27. 设遮热板温度为t_2'，则有：

$$\alpha_2 (t_f - t_{2'}) = \frac{E_{b_2} - E_{b_{2'}}}{\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1}$$

$$\alpha_1 (t_f - t_1) = \varepsilon_1 (E_{b_1} - E_{b_{2'}})$$

联立上面两式凑得$t_{2'}$及t_f

$$\Delta = \left| \frac{t_f - t_1}{t_1} \right| \times 100\%$$　　结果略。

28—35 （略）

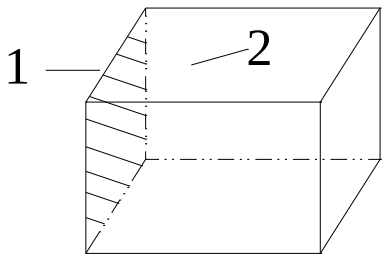
第十章　传热和换热器

1—7（略）

8. 已知：$F = 6 \times 4m^2, h = 4m, \varepsilon = 0.87, t_w = 40℃, t_f = 16℃$

求：$Q, Q_R/Q, Q_C/Q$

解：$Q_R = \dfrac{E_{b_1} - E_{b_2}}{\dfrac{1-\varepsilon}{\varepsilon F_1} + \dfrac{1}{F_1 Z_{1,2}} + \dfrac{1-\varepsilon}{\varepsilon F_2}} = 2.12kw$



z

1,2

=1

$Q_c = \alpha F_1 (t_w - t_f)$

而：按 $t_m = \dfrac{1}{2}(t_w + t_f) = 28℃$ 查表有：

$\beta_m = 3.31\times10^{-3}\ 1/K, \nu_m = 16\times10^{-6}\ m^2/s, \mathrm{Pr}_m = 0.712, \lambda_m = 2.64\times10^{-2}\ w/(m\cdot K)$

$(Gr\,\mathrm{Pr})_m = \dfrac{g\beta_m \Delta t h^3}{\nu_m^2}\mathrm{Pr}_m = 1.39\times10^{11}$

查表知：$c = 0.1, n = 1/3$

$\therefore \alpha = \dfrac{\lambda_m}{h} c\ (Gr\,\mathrm{Pr})_m^n = 3.42\ w/(m^2\cdot K)$

$Q_c = \alpha F(t_w - t_f) = 1.31kw$

$Q = Q_R + Q_c = 3.43kw$

$Q_R/Q = 61.8\%$，$Q_c/Q = 38.2\%$

9. 略

10. 已知：$\delta = 0.2m, \lambda_w = 0.6\ w/(m\cdot K), \varepsilon = 0.9, t_{f_1} = t_w = 18℃, t_F = -60℃$

$$h_1 = 0.592\ {}^{w}\!/_{(m^2 \cdot K)}, h_2 = 21.1\ {}^{w}\!/_{(m^2 \cdot K)}, t_{w_2} = 0℃,\ F_w >> F_{w_1}$$

求：$t_{w_1}, h_1^{'}, h_2^{'}, k$

解：∵ $F_w >> F_{w_1}$，且稳定时（热平衡）

$$\frac{\lambda_w}{\delta}(t_{w_1} - t_{w_1}) = h_1(t_{f_1} - t_{w_1}) + \varepsilon(E_{bw} - E_{bw_1})$$

带入数值并整理有：

11.略

12.略

13.已知：$t_1^{'} = 300℃$，$t_1^{''} = 180℃$，$t_2^{'} = 20℃$，$t_2^{''} = 150℃$

求：$\Delta t_{m\,顺}$ 、$\Delta t_{m\,逆}$ 、$\Delta t_{m\,交}$

解：（1）$\Delta t_{m\,顺} = \dfrac{\Delta t_{\max} - \Delta t_{\min}}{\ln \dfrac{\Delta t_{\max}}{\Delta t_{\min}}} = 111.93℃$

（2）同理　$\Delta t_{m\,逆} = 154.95℃$

（3）$P = \dfrac{t_2^{''} - t_2^{'}}{t_1^{'} - t_2^{'}} = 0.464, R = \dfrac{t_1^{'} - t_1^{''}}{t_2^{''} - t_2^{'}} = 0.923$

查图得：$\psi = 0.93$

∴ $\Delta t_{m\,交} = \psi\ \Delta t_{m\,逆} = 144.1℃$

14. 已知：$G_1 = 0.1\ {}^{kg}\!/_{s}, Cp_1 = 2.1\ {}^{kJ}\!/_{(kg \cdot K)}, t_1^{'} = 350℃, G_2 = G_1$

$$Cp_2 = 4.18\ {}^{kJ}\!/_{(kg \cdot K)}, t_2^{'} = 100℃, t_2^{''} = 200℃$$

（1）$k_1 = 500\ {}^{w}\!/_{(m^2 \cdot k)}, F_1 = 0.8m^2$

（2）$k_2 = 400\ {}^{w}\!/_{(m^2 \cdot k)}, F_2 = 1.2m^2$

求：$F_1^{'}, F_2^{'}$

解：由热平衡方程

$$Q = GCp_1\Delta t_1 = G_2Cp_2\Delta t_2, G_1 = G_2$$

知：$t_1^{''} = t_1^{'} - \dfrac{Cp_2}{Cp_1}\Delta t_1 = 144.62℃$

由此知（$t_2'' > t_1''$）此换热器需按逆流布置

有：$\Delta t_m = \dfrac{\Delta t_{\max} - \Delta t_{\min}}{\ln \dfrac{\Delta t_{\max}}{\Delta t_{\min}}} = 86.91℃$

$F_1' = \dfrac{Q_1}{k_1 \Delta t_m} = \dfrac{G_1 Cp_1 \Delta t_1}{k_1 \Delta t_m} = 0.99m^2 > 0.8 = F_1$，不适用

$F_2' = \dfrac{G_1 Cp_1 \Delta t_1}{k_2 \Delta t_m} = 1.2m^2 = F_2$，适用

15.已知：$t_1' = 77℃, t_2'' = 47℃, G_1 = 1 {kg}/{s}, Cp_1 = 1758 {J}/{(kg \cdot K)}, t_2' = 13℃$

$G_2 = 0.63 {kg}/{s}, k = 310 {w}/{(m^2 \cdot K)}$

求：$(1)F_{逆}(2)F_{1-2}(3)F_{交}$

解：由热平衡方程得：

$G_1 Cp_1 \Delta t_1 = G_2 Cp_2 \Delta t_2$

$t_2'' = t_2' + \dfrac{G_1 Cp_1}{G_2 Cp_2} \Delta t_1 = 33℃$

（1）逆流时 $\Delta t_{m逆} = \dfrac{\Delta t_{\max} - \Delta t_{\min}}{\ln \dfrac{\Delta t_{\max}}{\Delta t_{\min}}} = 38.79℃$

$F_{逆} = \dfrac{Q}{k \Delta t_m} = 4.39m^2$

（2）1-2 型

$p = \dfrac{t_2'' - t_2'}{t_1' - t_2'} = 0.313, R = \dfrac{t_1' - t_1''}{t_2'' - t_2'} = 1.5$

查图得：$\psi_{1-2} = 0.95$，知：

$\Delta t_{m1-2} = \psi \Delta t_{m逆} = 36.85℃$

$F_{1-2} = \dfrac{Q}{k \Delta t_{m1-2}} = 4.62m^2$

（3）同理得：$\psi_{交} = 0.94$，（$p = 0.313, R = 1.5$）

$\Delta t_{m\,交}=\psi_{交}\Delta t_{m\,逆}=36.46℃$

$F_{交}Q\div k\Delta t_{m\,交}=4.67m^2$

16.查标准状态（0℃,1*atm*）时的 $\rho=1.293\,kg/m^3$

即：$G_2=\rho\overline{v_2}=1.293\times\dfrac{1000}{60}=21.55\,kg/s$

其它计算同前（略）

17.略

18.已知: $G_2=0.125\,kg/s$, $Cp_2=4.2\,kg/(kg\cdot K)$, $t_2^{'}=40℃$, $t_2^{''}=95℃$

$G_1=0.125\,kg/s$, $Cp_1=2.1\,kg/(kg\cdot K)$, $t_1^{'}=210℃$

求：（1）$Q_{\max}$ （2）ε （3）$KF_{\min}$ （4）$KF_{\max}/KF_{\min}$

解：（1）$G_1Cp_1<G_2Cp_2$ 即 $(GCp)_{\min}=G_1Cp_1$

$Q_{\max}=(GCp)_{\min}(t_1^{'}-t_2^{'})=G_1Cp_1(t_1^{'}-t_2^{'})=44.63kw$

（2）$\varepsilon=\dfrac{(t^{'}-t^{''})_{\max}}{t_1^{'}-t_2^{'}}=\dfrac{t_1^{'}-t_1^{''}}{t_1^{'}-t_2^{'}}=0.647$

故：$t_1^{''}=t_1^{'}-\dfrac{G_2Cp_2}{G_1Cp_1}\Delta t_2=100℃$

$\Delta t_1=110℃>\Delta t_2=55℃$

（3）$Q=\varepsilon(GCp)_{\min}(t_1^{'}-t_2^{'})=57.74kw$

顺流时：$\Delta t_{m\,顺}=\dfrac{\Delta t_{\max}-\Delta t_{\min}}{\ln\dfrac{\Delta t_{\max}}{\Delta t_{\min}}}=46.79℃$

$KF_{顺}=\dfrac{Q}{t_{m顺}}=1.234kw\cdot℃^{-1}$

逆流时：$\Delta t_{m\,逆}=\dfrac{\Delta t_{\max}-\Delta t_{\min}}{\ln\dfrac{\Delta t_{\max}}{\Delta t_{\min}}}=84.54℃$

$KF_{逆}=\dfrac{Q}{t_{m逆}}=0.683kw\cdot℃^{-1}$

$\therefore KF_{\min} = KF_{逆} = 0.683\ kw\cdot ℃^{-1}$

$$\frac{KF_{\max}}{KF_{\min}} = \frac{KF_{逆}}{KF_{顺}} = 1.81$$

19.已知： $t_1^{'} = 300\ ℃$、$t_1^{''} = 100\ ℃$、$G_1 = 1\ {kg}/{s}$、$Cp_1 = 1\ {kJ}/{(kg\cdot K)}$

$t_2^{'} = 35\ ℃$、$t_2^{''} = 125\ ℃$、$K_0 = 100\ {w}/{(m^2\cdot K)}$

求： F

解：由热平衡方程式有：

$$G_2\ Cp_2 = G_1 Cp_1 \frac{t_1^{'} - t_1^{''}}{t_2^{''} - t_2^{'}} = 2.22\ kw\cdot ℃^{-1}$$

即有： $\dfrac{(GCp)_{\min}}{(GCp)_{\max}} = \dfrac{G_1 Cp_1}{G_2 Cp_2} = 0.45$

再： $\varepsilon = \dfrac{(t^{'} - t^{''})}{t_1^{'} - t_2^{'}} = 0.755$

查图得：NTU $= 2.5$

即：NTU $= \dfrac{KF}{(GCp)_{\min}} = \dfrac{KF}{G_1 Cp_1}$

$$F = \frac{\mathrm{NTU} G_1 Cp_1}{K} = 25 m^2$$

20.略（同上题）

21.已知： $p = 0.8\times 10^5\ pa$、$G_1 = 1.5\times 10^{-2}\ {kg}/{s}$、$t_2^{'} = 10\ ℃$、$t_2^{''} = 60\ ℃$

$K = 2000\ {w}/{(m^2\cdot K)}$、$t_{2a}^{'} = 30\ ℃$

求： F、G_{2a}

解：按 $p = 0.8\times 10^5\ pa$ **查表得：**

$t_s = 93.51\ ℃$、$r = 2274.3\ {kJ}/{kg}$

$$\Delta t_{\max} = \frac{\Delta t_{\max} - \Delta t_{\min}}{\ln \dfrac{\Delta t_{\max}}{\Delta t_{\min}}} = 54.76\ ℃$$

由 $Q = G_1 r = KF\Delta t_m$ **知**

$$F=\frac{G_1 r}{K\Delta t_m}=0.311m^2$$

再：$t_2^{'}\uparrow\rightarrow t_{2a}^{'}=30$　但$Q=cons$ 则有：

$$Q=G_1 r=G_2 Cp_2\Delta t_2=G_{2a}Cp_{2a}\Delta t_{2a}$$

取$Cp_2\approx Cp_{2a}$

$$\therefore\frac{G_{2a}}{G_2}=\frac{\Delta t_2}{\Delta t_{2a}}=\frac{t_2^{''}-t_2^{'}}{t_2^{''}-t_{2a}^{'}}=1.67$$

$$\therefore G_{2a}=1.67G_2$$　G_{2a}较G_2增大 **67%**

22. 已知：$t_1^{'}=100℃$、$t_1^{''}=60℃$、$t_2^{'}=20℃$、$t_2^{''}=50℃$

$$Q=2.5\times10^4 w、K=350\ {}^{w}\!/_{(m^2\cdot K)}、R_f=0.004\ {}^{m^2\cdot K}\!/_{w}$$

求：Q_a、$t_{1a}^{''}$、$t_{2a}^{''}$、F

解：$\Delta t_{\max}=\dfrac{\Delta t_{\max}-\Delta t_{\min}}{\ln\dfrac{\Delta t_{\max}}{\Delta t_{\min}}}=44.81℃$

$$\therefore F=\frac{Q}{K\Delta t_m}=1.594m^2$$

再：$G_1\ Cp_1=\dfrac{Q}{\Delta t_1}=625w\cdot℃^{-1}$

$$G_2\ Cp_2=\frac{Q}{\Delta t_2}=833.3w\cdot℃^{-1}$$

即：$\dfrac{(GCp)_{\min}}{(GCp)_{\max}}=\dfrac{G_1Cp_1}{G_2Cp_2}=0.75$

另：$\dfrac{1}{K_a}=\dfrac{1}{K}+R_f\Rightarrow K_a=(\dfrac{1}{K}+R_f)^{-1}=145.83\ {}^{w}\!/_{(m^2\cdot K)}$

$$\mathbf{NTU}=\frac{K_aF}{(GCp)_{\min}}=\frac{K_aF}{G_1Cp_1}=0.372$$

查图有：$\varepsilon=0.31$

$$\therefore Q_a=\varepsilon(GCp)_{\min}(t_1^{'}-t_2^{'})=\varepsilon(G_1Cp_1)(t_1^{'}-t_2^{'})=1.55\times10^4 w$$

$$t''_{1a} = t'_1 - \frac{Q_a}{G_1 Cp_1} = 75.2\ ℃$$

$$t''_{2a} = t'_{2a} - \frac{Q_a}{G_2 Cp_2} = 38.6\ ℃$$